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Paper 1 of 7 — PHY 101 A POWER POINT PRESENTATION

Level 100 • First Semester • Handout

3080 OSUN STATE UNIVERSITY, OSOGBO
COLLEGE OF SCIENCE, ENGINEERING AND TECHNOLOGY
FACULTY OF BASIC SCIENCE
DEPARTMENT OF PHYSICS
PHYSICS 101

Lecture Note Prepared

By

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COURSE CONTENT AREA TO COVER INCLUDE:

Space and Time

Units and Dimensions

Vectors and Scalars,

Differentiation of Vectors (Displacement, Velocity and Acceleration)

Physical Quantity and Dimensions

- When we say quantity in the literary sense of it, it may mean an amount or a number of something. By amount, we use words such as large, small, enormous, vast, huge, sufficient, and limited to describe a quantity. For example, we can say the material is available in sufficient or limited quantity.
- In science, quantity is defined either by specifying how it is measured or by stating how it is calculated.
- Quantity is the property of things which can be measured. The term also bring to mind the questions such as what is the distance of your friend's house to your school, how big is the fish or yam and what is the time it will take you to walk from your house to your friend's house.
- You can also calculate the speed of an athlete by dividing the total distance covered by the time of running.
- In Physics, physical quantity is used to describe any quantity that we might come across during our studying the subject. There are two types of physical quantity that can be distinguished
 - (a) Base quantities and
 - (b) Derived quantities.
- Base Quantities are those that were defined through measurement process. The measurement of these quantities is independent of each other and all other quantities can be derived from them. The measurement of base quantities should be accurate and precise. The reason for this is because all other quantities are expressed as algebraic combination of the base quantities.
- Base on this, the International Standards Organization recommends seven based

Quantities which form the international system of Quantities (ISQ).

- These ISQ base quantities and their symbols are listed in table 1 (a)

Table 1: Fundamental Quantity and their Symbol

S/N Fundamental quantity Symbol Fundamental

1 Mass M Kilogramme(kg)

2 Length L Metre(m)

3 Time T Second (s)

4 Angle θ Radian (rad)

5 Amount of substance N Mole (mol.)

6 Thermodynamic temperature T Kelvin (k)

7 Electric current I Ampere (A)

8 Luminous intensity E Candela (cd)

Units

- Measurements of physical quantities are expressed in terms of units, which are standardized values. For example, an athlete in a competition would cover length of a race, which is a physical quantity, can be expressed in units of meters (for sprinters) or kilometers (for distance runners). Without standardized units, it would be extremely difficult for scientists to express and compare measured values in a meaningful way.
- There are two major systems of units used in the world: SI units (also known as the metric system) and English units (also known as the customary or imperial system). English units were historically used in nations once ruled by the British Empire and are still widely used in the United States. Virtually every other country in the world now uses SI units as the standard; the metric system is also the standard system agreed upon by scientists and mathematicians. The acronym "SI" is derived from the French *Système International*. The units of these fundamental quantities are called fundamental units. These fundamental quantities, their symbols and units are listed in Table 1.1

Derived quantities

- As the name implies they are derived/obtained from the base quantities. Their derivation is as a result of calculations done on base quantities.
- These calculations can be multiplication of two or more base quantities together or the division of two base quantities or it can be combinations of both operators.
- The process of derivation can also be by multiplying or dividing one derived and a base quantity.
- Examples of derived quantities, their derivation processes and unit are listed in Table 1.2. The unit of the derived quantity is called derived unit.

Table 2: Derived quantities, derivation processes and their units

SN Derived Quantity Derivation Process Derived unit Symbol

- 1 Area Length x Breadth (L x B) meter squared m^2
 - 2 Volume Length x Breadth x Height meter cube m^3
 - 3 Velocity Displacement/Time meter per second m/s or ms^{-1}
 - 4 Acceleration Increase in velocity/time meter per second squared ms^{-2} or m/s^2
 - 5 Density Mass / Volume kilogram / meter cube kg/m^3 or kgm^{-3}
 - 6 Period (T) Time / Number of t/n s
- Oscillation
- 7 Force Mass x Acceleration kgm/s^2 N
 - 8 Resistance Potential / current V/A ohm(Ω)
 - 9 Potential Difference Electrical Energy/ Charge J / C V
 - 10 Pressure Force / Area N / m^2 or $N m^{-2}$ Pascal

Metric Prefixes

- SI units are part of the metric system. The metric system is convenient for scientific and engineering calculations because the units are categorized by factors of 10. Table 1.3 gives metric prefixes and symbols used to denote various factors of 10.
- Metric systems have the advantage that conversions of units involve only powers of 10. There are 100 centimeters in a meter, 1000 meters in a kilometer, and so on.
- In nonmetric systems, such as the system of U.S. customary units, the relationships are not as simple—there are 12 inches in a foot, 5280 feet in a mile, and so on.
- Another advantage of the metric system is that the same unit can be used over extremely large ranges of values simply by using an appropriate metric prefix. For example, distances in meters are suitable in construction, while distances in kilometers are appropriate for air travel, and the tiny measure of nanometers are convenient in optical design. With the metric system there is no need to invent new units for particular applications.

Table 3: Metric Prefixes for Powers of 10 and their Symbols

Prefix Symbol Value

yotta-	Y	10^{24}	
zetta-	Z	10^{21}	
exa	E	10^{18}	exameter Em
peta	P	10^{15}	petasecond Ps
tera	T	10^{12}	terawatt TW
giga	G	10^9	gigahertz GHz
mega	M	10^6	megacurie MCi
kilo	k	10^3	kilometer km
hecto	h	10^2	hectoliter hL
deka	da	10^1	dekagram dag

Table 4: Metric Prefixes for sub multiples of 10 and their Symbols

Prefix	Symbol	Value
yocto-	y	10^{-24}
zepto-	z	10^{-21}
atto	a	10^{-18} attosecond
femto	f	10^{-15} femtometre
pico	p	10^{-12} picofarad
nano	n	10^{-9} nanogram
micro	μ	10^{-6} micrometer
milli	m	10^{-3} millimeter
centi	c	10^{-2} centimeter
deci	d	10^{-1} deciliter

Dimensional Analysis

- The dimension of any physical quantity expresses its dependence on the base quantities as a product of symbols (or powers of symbols) representing the base quantities.
- Table 5 lists the base quantities and the symbols used for their dimension.
- For example, a measurement of length is said to have dimension L or L1, a measurement of mass has dimension M or M1, and a measurement of time has dimension T or T1.
- Like units, dimensions obey the rules of algebra. Thus, area is the product of two lengths and so has dimension L2, or length squared.
- Similarly, volume is the product of three lengths and has dimension L3, or length cubed.
- Speed has dimension length over time, L/T or LT^{-1} .
- Volumetric mass density has dimension M/L3 or ML^{-3} , or mass over length cubed.
- In general, the dimension of any physical quantity can be written as $L^a M^b T^c I^d \Theta^e N^f J^g$ for some powers a, b, c, d, e, f, and g. We can write the dimensions of a length in this form with and the remaining six powers all set equal to zero:
- Any quantity with a dimension that can be written so that all seven powers are zero (that is, its dimension is 1) is called dimensionless (or sometimes “of dimension 1,” because anything raised to the zero power is one). Physicists often call dimensionless quantities pure numbers.

Table 5: Base Quantities and Their Dimensions

- Physicists often use square brackets around the symbol for a physical quantity to represent the dimensions of that quantity. For example, if r is the radius of a cylinder and h is its height, then we write $[r] = L$ and $[h] = L$ to indicate the dimensions of the radius and height are both those of length, or L. Similarly, if we use the symbol A for the surface area of a cylinder and V for its volume, then $[A] = L^2$ and $[V] = L^3$. If we use the symbol m for the mass of the cylinder and ρ for the density of the material from which the cylinder is made, then $[m] = M$ and $[\rho] = ML^{-3}$.
- The importance of the concept of dimension arises from the fact that any mathematical equation relating physical quantities must be dimensionally consistent, which means the equation must obey the following rules:

- Every term in an expression must have the same dimensions; it does not make sense to add or subtract quantities of differing dimension (think of the old saying: “You can’t add apples and oranges”). In particular, the expressions on each side of the equality in an equation must have the same dimensions.
- The arguments of any of the standard mathematical functions such as trigonometric functions (such as sine and cosine), logarithms, or exponential functions that appear in the equation must be dimensionless.
- These functions require pure numbers as inputs and give pure numbers as outputs.
- If either of these rules is violated, an equation is not dimensionally consistent and cannot possibly be a correct statement of physical law. This simple fact can be used to check for typos or algebra mistakes, to help remember the various laws of physics, and even to suggest the form that new laws of physics might take. This last use of dimensions is beyond the scope of this text, but is something you will undoubtedly learn later in your academic career.

Using Dimensions to Remember an Equation

- Suppose we need the formula for the area of a circle for some computation.
- Two expressions may pop into our mind when we think of circles: and one expression is the circumference of a circle of radius r and the other is its area. But which is which?
- If either expression does not have the same dimensions as area, then it cannot possibly be the correct equation for the area of a circle

Solution

- We know the dimension of area is L^2 . Now, the dimension of the expression πr^2 is
- $[\pi r^2] = [\pi] \cdot [r^2] = 1 \cdot [L^2] = L^2$
- since the constant is a pure number and the radius is a length. Therefore, πr^2 has the dimension of area. Similarly, the dimension of the expression $2\pi r$ is
- $[2\pi r] = [2] \cdot [\pi] \cdot [r] = 1 \cdot 1 \cdot L = L$
- since the constants 2 and π are both dimensionless and the radius is a length. We see that has the dimension of length, which means it cannot possibly be an area.
- We rule out because it is not dimensionally consistent with being an area. We see that is dimensionally consistent with being an area, so if we have to choose between these two expressions, is the one to choose.

Checking Equations for Dimensional Consistency

- Consider the physical quantities s , v , a and t with dimensions $[s] = L$, $[v] = LT^{-1}$, $[a] = LT^{-2}$ and $[t] = T$. Determine whether each of the following equations is dimensionally consistent: (a) $s = ut + 0.5at^2$, (b) $s = ut^2 + 0.5t$ and (c) $v = \sin(at^2/s)$

Strategy

- By the definition of dimensional consistency, we need to check that each term in a given equation has the same dimensions as the other terms in that equation and that the arguments of any standard mathematical functions are dimensionless.

• Solution

(a.) There are no trigonometric, logarithmic, or exponential functions to worry about in this equation, so we need only look at

the dimensions of each term appearing in the equation. There are three terms, one in the left expression and two in the

expression on the right, so we look at each in turn:

- $[s] = L$ $[vt] = [v] \cdot [t] = LT^{-1} \cdot T = LT^0 = L$

- $[0.5at^2] = [a] \cdot [t^2] = LT^{-2} \cdot T^2 = LT^0 = L$

- All three terms have the same dimension, so this equation is dimensionally consistent.

(b.) Again, there are no trigonometric, exponential, or logarithmic functions, so we only need to look at the dimensions of each

of the three terms appearing in the equation:

- $[s] = L$ $[vt^2] = [v] \cdot [t^2] = LT^{-1} \cdot T^2 = LT$

- $[at] = [a] \cdot [t] = LT^{-2} \cdot T = LT^{-1}$

- None of the three terms has the same dimension as any other, so this is about as far from being dimensionally consistent as

you can get. The technical term for an equation like this is nonsense

- Dimensions are also used to indicate relationships among physical quantity. For example, the velocity V of a transverse wave along a stretched string depends on the

- Tension F in the string, Its length l and Its mass m . Find the actual equation

- The relationship can be written as

- (1.8)

- Where a , b and c are numbers we will find by dimension and k as a constant

- The dimension of velocity $v = LT^{-1}$, The dimension of tension $F = MLT^{-2}$

- The dimension of length (l) = L , The dimension of mass (m) = M

- From (1.8)

- $LT^{-1} = (MLT^{-2})^a L^b M^c$

- Equating indices of M , L and T on both sides

(1.9)

- $1 = a + b$ (1.10)

- $-1 = -2a$ (1.11)

(c.) This equation has a trigonometric function in it, so first we should check that the argument of the sine function is dimensionless:

- The argument is dimensionless. So far, so good. Now we need to check the dimensions of each of the two terms (that is, the left expression and the right expression) in the equation:

- $[v] = LT^{-1}$

- The two terms have different dimensions—meaning, the equation is not dimensionally consistent. This equation is another example of “nonsense.”
- One further point that needs to be mentioned is the effect of the operations of calculus on dimensions. We have seen that dimensions obey the rules of algebra, just like units, but what happens when we take the derivative of one physical quantity with respect to another or integrate a physical quantity over another? The derivative of a function is just the slope of the line tangent to its graph and slopes are ratios, so for physical quantities v and t , we have that the dimension of the derivative of v with respect to t is just the ratio of the dimension of v over that of t :
- Similarly, since integrals are just sums of products, the dimension of the integral of v with respect to t is simply the dimension of v times the dimension of t :
- By the same reasoning, analogous rules hold for the units of physical quantities derived from other quantities by integration or differentiation.

• VECTOR

• INTRODUCTION

Apart from classifying quantities into fundamental and derived quantities, quantities in physics can also be categorized into scalar quantity and vector quantity. Vectors are essential to physics and engineering. Many fundamental physical quantities are vectors, including displacement, velocity, force, and electric and magnetic vector fields. Scalar products of vectors define other fundamental scalar physical quantities, such as energy. Vector products of vectors define still other fundamental vector physical quantities, such as torque and angular momentum. In other words, vectors are a component part of physics in much the same way as sentences are a component part of literature.

• Scalars and Vectors

- Many familiar physical quantities can be specified completely by giving a single number and the appropriate unit. For example, “a class period lasts 50 min” or “the gas tank in my car holds 65 L” or “the distance between two posts is 100 m.” A physical quantity that can be specified completely in this manner is called a scalar quantity. Scalar is a synonym of “number.” Time, mass, distance, length, volume, temperature, and energy are examples of scalar quantities.
- Scalar quantities that have the same physical units can be added or subtracted according to the usual rules of algebra for numbers.

Similarly, a 60-cal serving of corn followed by a 200-cal serving of donuts gives off energy. When we multiply a scalar quantity by a number, we obtain the same scalar quantity but with a larger (or smaller) value. For example, if yesterday’s breakfast had 200 cal of energy and today’s breakfast has four times as much energy as it had yesterday, then today’s breakfast has 800 cal of energy. Two scalar quantities

can also be multiplied or divided by each other to form a derived scalar quantity. For example, if a train covers a distance of 100 km in 1.0 h, its speed is $100.0 \text{ km}/1.0 \text{ h} = 27.8 \text{ m/s}$, where the speed is a derived scalar quantity obtained by dividing distance by time

- Vectors
- Many physical quantities, however, cannot be described completely by just a single number of physical units. For example, when the U.S. Coast Guard dispatches a ship or a helicopter for a rescue mission, the rescue team must know not only the distance to the distress signal, but also the direction from which the signal is coming so they can get to its origin as quickly as possible.
- Physical quantities specified completely by giving a number of units (magnitude) and a direction are called vector quantities.
- Examples of vector quantities include displacement, velocity, position, force, and torque. In the language of mathematics, vector quantities are represented by mathematical objects called vectors (Figure 2.2).
- We can add or subtract two vectors, and we can multiply a vector by a scalar or by another vector, but we cannot divide by a vector. The operation of division by a vector is not defined.
- 1 Vector Addition
- The addition, subtraction and multiplication of vectors are not the same as those of ordinary number. For example, if a long – distance runner runs 15 km east, 7km west and finally 4 km north. The net displacement is not 26 km. The computation has to take direction into account. Because of this, and for convenience's sake, we need to represent a vector by a line and an arrow. The length here denotes the magnitude of the vector while the arrow on it indicates direction.
- Algebra Vectors in One Dimension
- The addition of two vectors which are not inclined to each other but are parallel or anti parallel is as shown in the fig. 6 below
- Algebra of Vectors in Two Dimensions
- When the two vectors are at angle 90° or orthogonal, we complete the square or rectangle, then their addition called the resultant is the diagonal that is drawn from where the two tails of the vector meet.
- The Two methods of adding or subtracting vectors in two dimensions are Geometrical and analytical methods
- Geometrical Method:
- This method can be by construction of a Triangle or a Parallelogram.
- To be able to do this, two given vectors A and B that are at a particular angle to each other, will be represented by two adjacent sides of a triangle inclined to each other at the angle between the two vectors.

- The third side R of the triangle is the sum or resultant of the two vectors.
 - This is a triangle way of adding two vectors.
 - The parallelogram method is to draw the two vectors by two sides of the parallelogram and then, we complete the parallelogram.
 - The line drawn from where the two tails of the vectors joined called the diagonal is then the addition or the resultant of the two vectors in question.
- The triangle and parallelogram way of addition gives the same answer for the resultant or sum of the two vectors
- If we have more than two vectors A, B. and C their addition by construction is as indicated in Fig. 1.3

Multiplication of Vectors

As mentioned earlier, derived quantity is formed when two or more quantities are multiplied together. For example, multiplication of distance by force gives work and again the multiplication of distance by force also give moment or torque. Certainly, there must be difference in these two multiplication operations. There are generally the

- (a) Multiplication of vector by a scalar will produce a vector
- (b) Multiplication of a vector by a vector will produce a scalar (dot product)
- (c) Multiplication of a vector by a vector will produce a vector (cross product).

(a) Multiplication of Vector by a Scalar

A good example of this operation is that of the product of mass and acceleration to give force. A vector quantity the direction is in the direction of the vector being multiplied with a scalar.

(b) Multiplication of a vector by another Vector

This product is called dot product of two vectors. Because the resultant of this multiplication is a scalar, it is also called Scalar product.

Dot or Scalar product is defined as

θ = the angle between the two vectors.

If are defined as $\vec{A}$ and $\vec{B}$ respectively, then

$$= (\vec{A} \cdot \vec{B})$$

$$= (AB \cos \theta)$$

Because =

Example: A force $F = 5\hat{i} + 3\hat{j}$ acts on a body and changes its position from $P_1 (4,5)$ to $P_2 (-3,6)$ in a plane. Find the work done on that body if the force is newton and displacement is in meter. The displacement $s = P_1 - P_2 = \hat{i} (-3 - 4) + \hat{j} (6-5) = -7\hat{i} + \hat{j}$

$$\therefore W =$$

$$= (5 -7) + (3 = -32J)$$

) Multiplication of vector by vector (cross) product

This multiplication is defined as

Where $\hat{n}$ is a unit vector per perpendicular to both .The magnitude of vector product is

The resultant of this multiplication is at right angle to both A and B. Vector product has the

property

If

Example: if $A = 2\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$ and $B = -\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}$. Find

$$= (-21 + 10) \mathbf{i}$$

$$= -11\mathbf{i} - 19\mathbf{j} - 7\mathbf{k}$$

// =

$$= 23.04$$

Coordinate Systems and Components of a Vector

Vectors are usually described in terms of their components in a coordinate system. In rectangular (Cartesian) xy plane, a point in a plane is described by a pair of coordinates (x, y). In like manner, a vector in a plane is described by a pair of its vector coordinates. The x -coordinate of vector A is called its x-component and the y coordinate is called the y component. The vector x component is a vector denoted by A_x . The y component is a vector denoted by A_y . In Cartesian system, the x and y components of a vector is the orthogonal projections of this vector onto the x- and y- axes respectively. In this way and using the parallelogram rule for vector addition, each vector in a Cartesian plane can be expressed as the vector sum of its vector components:

As shown in fig. 1.3 vector A is the diagonal of the rectangle where the x-component A_x is the side parallel to the x- axis and the y component A_y is the side parallel to the y-axis. Vector component A_x is orthogonal to vector component A_y

It is customary to denote the positive direction on the x-axis by the unit vector $\mathbf{i}$ and the positive direction on the y axis by the unit vector $\mathbf{j}$. Unit vectors of the axes, $\mathbf{i}$ and $\mathbf{j}$ define two orthogonal directions in the plane. From fig. 1.3, we can now write the x- and y- components of a vector in terms of the unit vector of the axes as

where A_x and A_y are numbers that define the vector components and are the scalar components of Vector A.

We obtain the component form of a vector

The scalar component of a vector by subtracting the origin point

coordinate from the end point coordinates i.e. $A_x = x_e - x_b$

and $A_y = y_e - y_b$

Example: The mouse of a computer is moved on the monitor from point (6.0cm, 1,6 cm) to an icon that is located at point (2.0cm, 4,5cm) What should be the displacement vector of the pointer?

Solution: $D_x = x_e - x_b = (2.0 - 6.0) \text{ cm} = -4.0\text{cm}$

$D_y = y_e - y_b = (4.5 - 1.6) \text{ cm} = +2.9\text{cm}$

The vector component form of the displacement vector is

$$D = D_x\mathbf{i} + D_y\mathbf{j} = (-4.0\text{cm})\mathbf{i} + (2.9\text{cm})\mathbf{j} = (-4.0\mathbf{i} + 2.9\mathbf{j}) \text{ cm}$$

We can find the magnitude of a vector when the scalar components are known. We use Pythagorean Theorem to find this magnitude as

$$A = \sqrt{A_x^2 + A_y^2} \quad A = \sqrt{A_x^2 + A_y^2}$$

This equation works if the scalar components are negatives. The direction of this magnitude is

obtained by finding the tangent function. So

A_y

$\tan \theta =$

A_x

This angle should be measured in the counterclockwise direction from the + x-axis to the vector.

From the example above the magnitude of the displacement vector $D = 4.9\text{cm}$ while its direction is -35.9° . This shows that vector D lies in the second quadrant, so its direction angle is

$$\theta_D = \theta + 180^\circ = -35.9^\circ + 180^\circ = 144.1^\circ$$

Vectors in Three Dimensions

An object in space is located by three coordinates (x, y, z) where coordinates x , and y specify locations in a plane, and coordinate z gives a vertical position above or below the plane. Three dimensional space has three orthogonal directions so we need not two but three unit vectors to define a three-dimensional coordinate system. In Cartesian coordinate system, the first two unit vectors are the unit vectors of x -axis i and the unit vector of the y -axis j . The third unit vector k is the direction of the z -axis. The order in which the unit vectors are labeled, which is the order in which the unit vectors appear is important because it defines the orientation of the coordinate system. The order x, y, z , which is equivalent to the order i, j, k , defines the standard right-handed coordinate system (positive orientation).

In three dimensional space, vector A has three vector components, the x -component $A_x = A_{xi}$, which is part of A along the x -axis, the y -component $A_y = A_{yj}$, which is part of A along the y -axis; and the z -component $A_z = A_{zk}$ which is part of A along the z -axis. A vector in three dimensional space is the vector sum of its three components

$$A = A_{xi} + A_{yj} + A_{zk}$$

The scalar component of the z -component is given by

$$A_z = z_e - z_b$$

If we know the beginning and the end coordinate $b(x_b, y_b, z_b)$ and $e(x_e, y_e, z_e)$ respectively since we have known that of A_x and A_y as discussed.

Magnitude A is obtained by generalizing to three dimensions

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2} \quad A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

- Mathematical Tools
- Introduction
- In physics, the knowledge of mathematical tool is necessary and important in order to treat and explain many concepts so that their meaning can be clearly explained. Such mathematical tools include derivatives (i.e. differentiation and integration) and trigonometric functions. These tools are only highlighted since their complete treatment is obtained in the relevant mathematics courses that we have offered or come across as we progress in our study of physics. So their treatment is beyond the scope of this course

1.1 Derivatives

When a graph of $y = 3x^2$ is plotted, the slope of this plotting when determine is not constant but

varies with the value of x . But the mathematician showed that it is possible to derive another function of x that will allow us to determine the slope at any value of x . The equation derived is called the derivative of the original equation. The process of deriving this equation is called

Differentiation and is represented as $\frac{dy}{dx}$. Mathematicians have shown that the first derivative of

the function $y = kx^n$

is

$\frac{dy}{dx}$

$= nkx^{n-1}$

1.1

$\frac{dy}{dx}$

$\frac{dy}{dx}$

For example, if $y = 6x^3$, then the derivative $= 18x^2$, while, $y = 3x^4 + 4x^3 + 2x^2 + 5x$, has the

$\frac{dy}{dx}$

$\frac{dy}{dx}$

derivative of $= 12x^3 + 12x^2 + 4x + 5$

$\frac{dy}{dx}$

Some common derivatives include

$\frac{dy}{dx}$

(i) $y = \cos x, \frac{dy}{dx} = -\sin x$ 1.2

$\frac{dy}{dx}$

$\frac{dy}{dx}$

(ii) $y = \sin x, \frac{dy}{dx} = \cos x$ 1.3

$\frac{dy}{dx}$

$\frac{dy}{dx}$

(iii) $y = A \cos bx, \frac{dy}{dx} = -Ab \sin bx$ 1.4

$\frac{dy}{dx}$

$\frac{dy}{dx}$

$y = A \sin bx, \frac{dy}{dx} = Ab \cos bx$ 1.5

$\frac{dy}{dx}$

Integration

This is the reverse of differentiation. Integration is denoted by the sign $\int$. Mathematicians have shown that

$\int b \, dx = bx + c$

1.6

$\int$

1.6

For example, $\int 12x^3 \, dx = 3x^4 + c$

A special integration is that of

1

$$\int \frac{1}{x} dx = \log a x + c \quad 17$$

The above is the general rule in which other common integration obey.

THANK YOU FOR LISTENING

Paper 2 of 7 — PHY101(Elasticity & Hydrostatic)-1

Level 100 • First Semester • Handout

3080

OSUN STATE UNIVERSITY

COLLEGE OF SCIENCE, ENGINEERING AND TECHNOLOGY

FACULTY OF BASIC AND APPLIED SCIENCES

DEPARTMENT OF PHYSICS

2024/2025 HARMATTAN SEMESTER

COURSE CODE: PHY 101 COURSE: GENERAL PHYSICS UNIT: 3

LECTURER: DR. Y.A AJAYEOBA

3080

Course Contents

- Elasticity
- Hydrostatics

3080 ELASTICITY

- Elasticity is the ability of a deformed material body to return to its original shape and size when the forces causing the deformation are removed. A body with this ability is said to behave (or respond) elastically.
- Most solid materials exhibit elastic behaviour, but there is a limit to the magnitude of the force and the accompanying deformation within which elastic recovery is possible for any given material. This limit is called the elastic limit

3080 ELASTICITY

- Elastic limit is the maximum stress or force per unit area within a solid material that can arise before the onset of permanent deformation. Stresses beyond the elastic limit cause a material to yield or flow.
- For such materials the elastic limit marks the end of elastic behaviour and the beginning of plastic behaviour. For most brittle materials, stresses beyond the elastic limit result in fracture with almost no plastic deformation.

3080 ELASTICITY

- Elastic deformation is when an object has a force applied but can still go back to its original length and shape after the force has been removed.

- Inelastic deformation is when the object with an applied force doesn't return to its original length and shape when the force is removed.

3080 ELASTICITY

- There is a relationship between the force applied to an elastic object and how much the object extends due to this force.
- The extension of an elastic object (i.e a spring) is directly proportional to the force applied to the object.
- Force = Spring constant $\times$ extension
- $F = kx$ ----- -1
- $k = \frac{\text{force}}{\text{extension}}$ (N/m)
- $x = \frac{\text{force}}{\text{spring constant}}$ (m)
- $E = \frac{\text{stress}}{\text{strain}}$ (N/m²)

HOOKE'S LAW

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- Hooke's law states that the strain of the material is proportional to the applied stress within the elastic limit of that material.
- When the elastic materials are stretched, the atoms and molecules deform until stress is applied, and when the stress is removed, they return to their initial state

- Stress $\propto$ Strain OR $\sigma = E \epsilon$ ----- -2

- E is constant of proportionality known as modulus of elasticity. It depends on the material being deformed and on the nature of the deformation.

3080 ELASTICITY

3080 ELASTICITY

3080 ELASTICITY

3080 ELASTICITY

3080 ELASTICITY

YOUNG'S MODULUS OF ELASTICITY

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- Young's modulus is a measure of the stiffness of an elastic material
- It is the ratio of longitudinal stress to linear strain.

$$\bullet \Delta =$$

0.000000 0.000000

0

$$\bullet \Delta = \text{-----} -3$$

0

Recall that:

0

$$\bullet \Delta = \text{-----} -4$$

0

Δ

$$\bullet \Delta = \text{-----} -5$$

0

YOUNG'S MODULUS OF ELASTICITY

3080

Therefore,

00

$$\bullet \Delta = \text{-----} -6$$

Δ

Where,

Y = Young's modulus

σ = Stress applied

ϵ = Strain related to the applied stress

F = Force exerted by the object

A = Actual cross-sectional area

ΔL = change in the length

L_0 = actual length

EXAMPLES

3080

1. A spring with $k=150$ N/m is compressed by 0.02 m. Calculate the force required.

$$F = kx$$

$$F = 150 \times 0.02$$

$$F = 3$$

2. A spring stretches by 0.05m when a 10N force is applied. Find the spring constant k

0

$$F =$$

0

10

$$k = \frac{F}{x}$$

0.05

3. A copper wire with Young's modulus 1.1×10^{11} N/m², length 3m, and cross-sectional area $4 \times$

10^{-6} m^2 elongates by 0.5mm. Calculate the force applied.

ΔL

$F =$

ΔL

ΔL

$F =$

F

$$(1.1 \times 10^{11}) \times (4 \times 10^{-6}) \times 0.0005$$

$$F = 73.3 \text{ N}$$

3

EXAMPLES

4. A wire of length 2m, cross-sectional area $3 \times 10^{-6} \text{ m}^2$, and Young's modulus 3080

$2.0 \times 10^{11} \text{ N/m}^2$ is subjected to a tensile force of 300N. Find the extension.

ΔL

$$\Delta L =$$

ΔL

$$300 \times 2$$

$$\Delta L = 11 \times 10^{-6}$$

$$= 11 \mu\text{m}$$

$$(2.0 \times 10) \times (3 \times 10)$$

5. A wire of radius 0.01m, original length of 2m is put under a tensile force of 75N. The wire extends by 1.5m. Calculate the tensile stress and strain of the wire.

σ

$$\text{Tensile stress } \sigma =$$

σ

$$75$$

$$\sigma = 238853.5 \text{ N/m}^2$$

$$1.5$$

$$\Delta L = 1.5$$

$$\text{Tensile strain } \epsilon = \frac{\Delta L}{L} = \frac{1.5}{2} = 0.75$$

$$\epsilon = 0.75$$

SHEAR MODULUS

$$3080$$

- Shear modulus, also known as Modulus of rigidity
- It is the measure of the rigidity of the body, given by the ratio of shear stress to shear strain. It is often denoted by G sometimes by S or μ .

τ Shear stress

$$\tau = \frac{F}{A}$$

Shear Modulus

•

• $\Delta L = \frac{FL}{A}$ ————— —7

• ΔL

- F is the force acting on the object.
- A is the area on which the force is acting.
- ΔL is the transverse displacement
- L is the initial length.

SHEAR MODULUS

3080 • The modulus of rigidity is the elastic coefficient when a shear force is applied resulting in lateral deformation. It gives us a measure of how rigid a body is.

• The following example will give you a clear understanding of how the shear modulus helps in defining the rigidity of any material.

- Shear modulus of wood is 6.2×10^8 Pa
- Shear modulus of steel is 7.2×10^{10} Pa
- Thus, it implies that steel is a lot more rigid than wood, around 127 times more!

EXAMPLES

3080

6. If the stress experienced by a body is $5 \times 10^4 \text{ N/m}^2$ and strain is 4×10^{-2} , what will be the Shear Modulus?

Shear Modulus

Shear Modulus =

Shear Modulus

4

5×10^4

$\Delta L = -2$

$= 1.25 \times 10^6 \text{ N/m}^2$

4×10^{-2}

7. The Force exerted on a body of length 20 m and area 200 m^2 is 5000 N. It undergoes a 1 m change in length due to the force being exerted. Find the shear modulus.

Shear Modulus

•

• ΔL

5000×20

•

200×1

$\Delta L = 5000 \times 10^{-2}$

8. The Force exerted on a body of length 10 m and area 100 m^2 is 400 N. It undergoes a 0.02 m

change in length due to the force being exerted. Find the shear modulus.

ΔL

ΔF

ΔA

400×10

$\Delta F = 2000 \times 10^{-2}$

100×0.02

BULK MODULUS

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- The bulk modulus is the measure of how resistant a material (solid or fluid) is to compression. It is defined as the ratio of pressure increase to the relative decrease of the material volume. It therefore represents how resistant a material is to changing volume under compression that is applied on all sides

- It is given by the ratio of pressure applied to the corresponding relative decrease in the volume of the material

BULK MODULUS

3080

ΔP

$\Delta V = \text{-----} -8$

$\Delta V/V$

Where:

- B: Bulk modulus
- ΔP : change of the pressure or force applied per unit area on the material
- ΔV : change of the volume of the material due to the compression
- V: Initial volume of the material

EXAMPLES

3080

9. Calculate the bulk modulus of a material that experienced a change of pressure of $5 \times 10^4 \text{ Pa}$ and its volume goes from 4 m^3 to 3.9 m^3

ΔP

ΔV

$\Delta V/V$

$5 \times 10^4 \text{ Pa}$ $1 \text{ m}^3 = 10^{-6} \text{ m}^3$

$(5 \times 10^4) \times 4 \times 10^{-6} \text{ Pa}$

$\Delta P = 2 \times 10$

$(4 - 3.9) \times 10^{-6}$

10. Using Bulk Modulus Formula, calculate the bulk modulus of a body that practices a change in stress of $4 \times 10^4 \text{ N m}^{-2}$ and its volume changes from 5 m^3 to 4.8 m^3 . (Ans: $106 \times 10^6 \text{ N m}^{-2}$)

11. A material experiences a pressure increase of $2 \times 10^5 \text{ N m}^{-2}$, and its volume changes from 1.0 m^3 to 0.98 m^3 . Calculate the bulk modulus. (Ans: $1 \times 10^7 \text{ N m}^{-2}$)

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HYDROSTATIC PRESSURE

- Hydrostatic pressure is defined as the pressure exerted by a fluid at equilibrium at any point of time due to the force of gravity.
- Hydrostatic pressure is proportional to the depth measured from the surface as the weight of the fluid increases when a downward force is applied.
- The fluid pressure can be caused by gravity, acceleration or forces when in a closed container.

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- Consider a layer of water from the top of the bottle. There is the pressure exerted by the layer of water acting on the sides of the bottle. As we move down from the top of the bottle to the bottom, the pressure exerted by the top layer on the bottom adds up. This phenomenon is responsible for more pressure at the bottom of the container.

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The formula used to calculate the hydrostatic pressure: $p = \rho gh$
where,

p is the pressure exerted by the liquid in Pascal

ρ is the density of the liquid in kg m^{-3}

g is the acceleration due to gravity taken as 9.81 m s^{-2}

h is the height of the fluid column in m

BUOYANCY

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- Buoyancy is the tendency of an object to float in a fluid. All liquids and gases in the presence of gravity exert an upward force known as the buoyant force on any object immersed in them.
- Buoyancy results from the differences in pressure acting on opposite sides of an object immersed in a static fluid.
- The buoyant force is the upward force exerted on an object wholly or partly immersed in a fluid. This upward force is also called Upthrust. Due to the buoyant

force, a body submerged partially or fully in a fluid appears to lose its weight, i.e. appears to be lighter

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FACTORS AFFECTING BUOYANT FORCE

1. The density of the fluid
2. The volume of the fluid displaced
3. The local acceleration due to gravity

- An object whose density is greater than that of the fluid in which it is submerged tends to sink. If the object is either less dense than the liquid or is shaped appropriately (as in a boat), the force can keep the object afloat. In terms of relative density, if the relative density is less than one, it floats in water and substances with a relative density greater than one sink in water.

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What Causes Buoyant Force?

- When an object is immersed in water or any other fluid, the object experiences a force from the downward direction opposite to the gravitational pull, which is responsible for the decrease in its weight. This upward force exerted by the fluid opposes the weight of an object immersed in a fluid. The pressure in a fluid column increases with depth. Thus, the pressure at the bottom of an object submerged in the fluid is greater than that at the top. The difference in this pressure results in a net upward force on the object, which we define as buoyancy

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ARCHIMEDES' PRINCIPLES

- Archimedes' principle states that:

"The upward buoyant force that is exerted on a body immersed in a fluid, whether partially or fully submerged, is equal to the weight of the fluid that the body displaces and acts in the upward direction at the center of mass of the displaced fluid"

- When an object is partially or fully immersed in a liquid, the apparent loss of weight is equal to the weight of the liquid displaced by it.

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- If you look at the figure, the weight due to gravity is opposed by the thrust provided by the fluid. The object inside the liquid only feels the total force acting on it as the weight. Because the actual gravitational force is decreased by the liquid's upthrust, the object feels as though its weight is reduced. The apparent weight is thus given by:

- Apparent weight = Weight of object (in the air) - Thrust force (buoyancy)

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Archimedes' Principle Formula

- In simple form, the Archimedes law states that the buoyant force on an object is equal to the weight of the fluid displaced by the object.

Mathematically written as:

$$F_b = \rho \times g \times V$$

Where

 F_b is the buoyant force, ρ is the density of the fluid, V is the submerged volume, and g is the acceleration due to gravity

Example:

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12. Calculate the resulting force, if a steel ball of radius 6 cm is immersed in water.

4 3

$$\text{Volume of sphere } V = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi (0.06)^3$$

3

4

$$V = \frac{4}{3} \pi (0.06)^3 = 0.000904 \text{ m}^3$$

3

$$F_b = \rho V g$$

$$F_b = 1000 \times 9.81 \times 0.000904$$

$$F_b = 8.87 \text{ N}$$

Example:

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13. A cubical block of wood of 1m side and density 500 kg m^{-3} floats in seawater of density 1000 kg m^{-3} . The depth of immersion of the block is (Ans: 0.5m)

Paper 3 of 7 — PHY 101 A POWER POINT PRESENTATION

Level 100 • First Semester • Handout

3080 OSUN STATE UNIVERSITY, OSOGBO
COLLEGE OF SCIENCE, ENGINEERING AND TECHNOLOGY
FACULTY OF BASIC SCIENCE
DEPARTMENT OF PHYSICS
PHYSICS 101

Lecture Note Prepared

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COURSE CONTENT AREA TO COVER INCLUDE:

Space and Time

Units and Dimensions

Vectors and Scalars,

Differentiation of Vectors (Displacement, Velocity and Acceleration)

Physical Quantity and Dimensions

- When we say quantity in the literary sense of it, it may mean an amount or a number of something. By amount, we use words such as large, small, enormous, vast, huge, sufficient, and limited to describe a quantity. For example, we can say the material is available in sufficient or limited quantity.
- In science, quantity is defined either by specifying how it is measured or by stating how it is calculated.
- Quantity is the property of things which can be measured. The term also bring to mind the questions such as what is the distance of your friend's house to your school, how big is the fish or yam and what is the time it will take you to walk from your house to your friend's house.
- You can also calculate the speed of an athlete by dividing the total distance covered by the time of running.
- In Physics, physical quantity is used to describe any quantity that we might come across during our studying the subject. There are two types of physical quantity that can be distinguished
 - (a) Base quantities and
 - (b) Derived quantities.
- Base Quantities are those that were defined through measurement process. The measurement of these quantities is independent of each other and all other quantities can be derived from them. The measurement of base quantities should be accurate and precise. The reason for this is because all other quantities are expressed as algebraic combination of the base quantities.
- Base on this, the International Standards Organization recommends seven based

Quantities which form the international system of Quantities (ISQ).

- These ISQ base quantities and their symbols are listed in table 1 (a)

Table 1: Fundamental Quantity and their Symbol

S/N Fundamental quantity Symbol Fundamental

1 Mass M Kilogramme(kg)

2 Length L Metre(m)

3 Time T Second (s)

4 Angle θ Radian (rad)

5 Amount of substance N Mole (mol.)

6 Thermodynamic temperature T Kelvin (k)

7 Electric current I Ampere (A)

8 Luminous intensity E Candela (cd)

Units

- Measurements of physical quantities are expressed in terms of units, which are standardized values. For example, an athlete in a competition would cover length of a race, which is a physical quantity, can be expressed in units of meters (for sprinters) or kilometers (for distance runners). Without standardized units, it would be extremely difficult for scientists to express and compare measured values in a meaningful way.
- There are two major systems of units used in the world: SI units (also known as the metric system) and English units (also known as the customary or imperial system). English units were historically used in nations once ruled by the British Empire and are still widely used in the United States. Virtually every other country in the world now uses SI units as the standard; the metric system is also the standard system agreed upon by scientists and mathematicians. The acronym "SI" is derived from the French *Système International*. The units of these fundamental quantities are called fundamental units. These fundamental quantities, their symbols and units are listed in Table 1.1

Derived quantities

- As the name implies they are derived/obtained from the base quantities. Their derivation is as a result of calculations done on base quantities.
- These calculations can be multiplication of two or more base quantities together or the division of two base quantities or it can be combinations of both operators.
- The process of derivation can also be by multiplying or dividing one derived and a base quantity.
- Examples of derived quantities, their derivation processes and unit are listed in Table 1.2. The unit of the derived quantity is called derived unit.

Table 2: Derived quantities, derivation processes and their units

SN Derived Quantity Derivation Process Derived unit Symbol

- 1 Area Length x Breadth (L x B) meter squared m^2
 - 2 Volume Length x Breadth x Height meter cube m^3
 - 3 Velocity Displacement/Time meter per second m/s or ms^{-1}
 - 4 Acceleration Increase in velocity/time meter per second squared ms^{-2} or m/s^2
 - 5 Density Mass / Volume kilogram / meter cube kg/m^3 or kgm^{-3}
 - 6 Period (T) Time / Number of t/n s
- Oscillation
- 7 Force Mass x Acceleration kgm/s^2 N
 - 8 Resistance Potential / current V/A ohm(Ω)
 - 9 Potential Difference Electrical Energy/ Charge J / C V
 - 10 Pressure Force / Area N / m^2 or $N m^{-2}$ Pascal

Metric Prefixes

- SI units are part of the metric system. The metric system is convenient for scientific and engineering calculations because the units are categorized by factors of 10. Table 1.3 gives metric prefixes and symbols used to denote various factors of 10.
- Metric systems have the advantage that conversions of units involve only powers of 10. There are 100 centimeters in a meter, 1000 meters in a kilometer, and so on.
- In nonmetric systems, such as the system of U.S. customary units, the relationships are not as simple—there are 12 inches in a foot, 5280 feet in a mile, and so on.
- Another advantage of the metric system is that the same unit can be used over extremely large ranges of values simply by using an appropriate metric prefix. For example, distances in meters are suitable in construction, while distances in kilometers are appropriate for air travel, and the tiny measure of nanometers are convenient in optical design. With the metric system there is no need to invent new units for particular applications.

Table 3: Metric Prefixes for Powers of 10 and their Symbols

Prefix Symbol Value

yotta-	Y	10^{24}	
zetta-	Z	10^{21}	
exa	E	10^{18}	exameter Em
peta	P	10^{15}	petasecond Ps
tera	T	10^{12}	terawatt TW
giga	G	10^9	gigahertz GHz
mega	M	10^6	megacurie MCi
kilo	k	10^3	kilometer km
hecto	h	10^2	hectoliter hL
deka	da	10^1	dekagram dag

Table 4: Metric Prefixes for sub multiples of 10 and their Symbols

Prefix	Symbol	Value
yocto-	y	10^{-24}
zepto-	z	10^{-21}
atto	a	10^{-18} attosecond
femto	f	10^{-15} femtometre
pico	p	10^{-12} picofarad
nano	n	10^{-9} nanogram
micro	μ	10^{-6} micrometer
milli	m	10^{-3} millimeter
centi	c	10^{-2} centimeter
deci	d	10^{-1} deciliter

Dimensional Analysis

- The dimension of any physical quantity expresses its dependence on the base quantities as a product of symbols (or powers of symbols) representing the base quantities.
- Table 5 lists the base quantities and the symbols used for their dimension.
- For example, a measurement of length is said to have dimension L or L1, a measurement of mass has dimension M or M1, and a measurement of time has dimension T or T1.
- Like units, dimensions obey the rules of algebra. Thus, area is the product of two lengths and so has dimension L2, or length squared.
- Similarly, volume is the product of three lengths and has dimension L3, or length cubed.
- Speed has dimension length over time, L/T or LT^{-1} .
- Volumetric mass density has dimension M/L3 or ML^{-3} , or mass over length cubed.
- In general, the dimension of any physical quantity can be written as $L^a M^b T^c I^d \Theta^e N^f J^g$ for some powers a, b, c, d, e, f, and g. We can write the dimensions of a length in this form with and the remaining six powers all set equal to zero:
- Any quantity with a dimension that can be written so that all seven powers are zero (that is, its dimension is 1) is called dimensionless (or sometimes “of dimension 1,” because anything raised to the zero power is one). Physicists often call dimensionless quantities pure numbers.

Table 5: Base Quantities and Their Dimensions

- Physicists often use square brackets around the symbol for a physical quantity to represent the dimensions of that quantity. For example, if r is the radius of a cylinder and h is its height, then we write $[r] = L$ and $[h] = L$ to indicate the dimensions of the radius and height are both those of length, or L. Similarly, if we use the symbol A for the surface area of a cylinder and V for its volume, then $[A] = L^2$ and $[V] = L^3$. If we use the symbol m for the mass of the cylinder and ρ for the density of the material from which the cylinder is made, then $[m] = M$ and $[\rho] = ML^{-3}$.
- The importance of the concept of dimension arises from the fact that any mathematical equation relating physical quantities must be dimensionally consistent, which means the equation must obey the following rules:

- Every term in an expression must have the same dimensions; it does not make sense to add or subtract quantities of differing dimension (think of the old saying: “You can’t add apples and oranges”). In particular, the expressions on each side of the equality in an equation must have the same dimensions.
- The arguments of any of the standard mathematical functions such as trigonometric functions (such as sine and cosine), logarithms, or exponential functions that appear in the equation must be dimensionless.
- These functions require pure numbers as inputs and give pure numbers as outputs.
- If either of these rules is violated, an equation is not dimensionally consistent and cannot possibly be a correct statement of physical law. This simple fact can be used to check for typos or algebra mistakes, to help remember the various laws of physics, and even to suggest the form that new laws of physics might take. This last use of dimensions is beyond the scope of this text, but is something you will undoubtedly learn later in your academic career.

Using Dimensions to Remember an Equation

- Suppose we need the formula for the area of a circle for some computation.
- Two expressions may pop into our mind when we think of circles: and one expression is the circumference of a circle of radius r and the other is its area. But which is which?
- If either expression does not have the same dimensions as area, then it cannot possibly be the correct equation for the area of a circle

Solution

- We know the dimension of area is L^2 . Now, the dimension of the expression πr^2 is
- $[\pi r^2] = [\pi] \cdot [r^2] = 1 \cdot [L^2] = L^2$
- since the constant is a pure number and the radius is a length. Therefore, πr^2 has the dimension of area. Similarly, the dimension of the expression $2\pi r$ is
- $[2\pi r] = [2] \cdot [\pi] \cdot [r] = 1 \cdot 1 \cdot L = L$
- since the constants 2 and π are both dimensionless and the radius is a length. We see that has the dimension of length, which means it cannot possibly be an area.
- We rule out because it is not dimensionally consistent with being an area. We see that is dimensionally consistent with being an area, so if we have to choose between these two expressions, is the one to choose.

Checking Equations for Dimensional Consistency

- Consider the physical quantities s , v , a and t with dimensions $[s] = L$, $[v] = LT^{-1}$, $[a] = LT^{-2}$ and $[t] = T$. Determine whether each of the following equations is dimensionally consistent: (a) $s = ut + 0.5at^2$, (b) $s = ut^2 + 0.5t$ and (c) $v = \sin(at^2/s)$

Strategy

- By the definition of dimensional consistency, we need to check that each term in a given equation has the same dimensions as the other terms in that equation and that the arguments of any standard mathematical functions are dimensionless.

• Solution

(a.) There are no trigonometric, logarithmic, or exponential functions to worry about in this equation, so we need only look at

the dimensions of each term appearing in the equation. There are three terms, one in the left expression and two in the

expression on the right, so we look at each in turn:

- $[s] = L$ $[vt] = [v] \cdot [t] = LT^{-1} \cdot T = LT^0 = L$

- $[0.5at^2] = [a] \cdot [t^2] = LT^{-2} \cdot T^2 = LT^0 = L$

- All three terms have the same dimension, so this equation is dimensionally consistent.

(b.) Again, there are no trigonometric, exponential, or logarithmic functions, so we only need to look at the dimensions of each

of the three terms appearing in the equation:

- $[s] = L$ $[vt^2] = [v] \cdot [t^2] = LT^{-1} \cdot T^2 = LT$

- $[at] = [a] \cdot [t] = LT^{-2} \cdot T = LT^{-1}$

- None of the three terms has the same dimension as any other, so this is about as far from being dimensionally consistent as

you can get. The technical term for an equation like this is nonsense

- Dimensions are also used to indicate relationships among physical quantity. For example, the velocity V of a transverse wave along a stretched string depends on the

- Tension F in the string, Its length l and Its mass m . Find the actual equation

- The relationship can be written as

- (1.8)

- Where a , b and c are numbers we will find by dimension and k as a constant

- The dimension of velocity $v = LT^{-1}$, The dimension of tension $F = MLT^{-2}$

- The dimension of length (l) = L , The dimension of mass (m) = M

- From (1.8)

- $LT^{-1} = (MLT^{-2})^a L^b M^c$

- Equating indices of M , L and T on both sides

(1.9)

- $1 = a + b$ (1.10)

- $-1 = -2a$ (1.11)

(c.) This equation has a trigonometric function in it, so first we should check that the argument of the sine function is dimensionless:

- The argument is dimensionless. So far, so good. Now we need to check the dimensions of each of the two terms (that is, the left expression and the right expression) in the equation:

- $[v] = LT^{-1}$

- The two terms have different dimensions—meaning, the equation is not dimensionally consistent. This equation is another example of “nonsense.”
- One further point that needs to be mentioned is the effect of the operations of calculus on dimensions. We have seen that dimensions obey the rules of algebra, just like units, but what happens when we take the derivative of one physical quantity with respect to another or integrate a physical quantity over another? The derivative of a function is just the slope of the line tangent to its graph and slopes are ratios, so for physical quantities v and t , we have that the dimension of the derivative of v with respect to t is just the ratio of the dimension of v over that of t :
- Similarly, since integrals are just sums of products, the dimension of the integral of v with respect to t is simply the dimension of v times the dimension of t :
- By the same reasoning, analogous rules hold for the units of physical quantities derived from other quantities by integration or differentiation.

• VECTOR

• INTRODUCTION

Apart from classifying quantities into fundamental and derived quantities, quantities in physics can also be categorized into scalar quantity and vector quantity. Vectors are essential to physics and engineering. Many fundamental physical quantities are vectors, including displacement, velocity, force, and electric and magnetic vector fields. Scalar products of vectors define other fundamental scalar physical quantities, such as energy. Vector products of vectors define still other fundamental vector physical quantities, such as torque and angular momentum. In other words, vectors are a component part of physics in much the same way as sentences are a component part of literature.

• Scalars and Vectors

- Many familiar physical quantities can be specified completely by giving a single number and the appropriate unit. For example, “a class period lasts 50 min” or “the gas tank in my car holds 65 L” or “the distance between two posts is 100 m.” A physical quantity that can be specified completely in this manner is called a scalar quantity. Scalar is a synonym of “number.” Time, mass, distance, length, volume, temperature, and energy are examples of scalar quantities.
- Scalar quantities that have the same physical units can be added or subtracted according to the usual rules of algebra for numbers.

Similarly, a 60-cal serving of corn followed by a 200-cal serving of donuts gives off energy. When we multiply a scalar quantity by a number, we obtain the same scalar quantity but with a larger (or smaller) value. For example, if yesterday’s breakfast had 200 cal of energy and today’s breakfast has four times as much energy as it had yesterday, then today’s breakfast has 800 cal of energy. Two scalar quantities

can also be multiplied or divided by each other to form a derived scalar quantity. For example, if a train covers a distance of 100 km in 1.0 h, its speed is $100.0 \text{ km}/1.0 \text{ h} = 27.8 \text{ m/s}$, where the speed is a derived scalar quantity obtained by dividing distance by time

- Vectors
- Many physical quantities, however, cannot be described completely by just a single number of physical units. For example, when the U.S. Coast Guard dispatches a ship or a helicopter for a rescue mission, the rescue team must know not only the distance to the distress signal, but also the direction from which the signal is coming so they can get to its origin as quickly as possible.
- Physical quantities specified completely by giving a number of units (magnitude) and a direction are called vector quantities.
- Examples of vector quantities include displacement, velocity, position, force, and torque. In the language of mathematics, vector quantities are represented by mathematical objects called vectors (Figure 2.2).
- We can add or subtract two vectors, and we can multiply a vector by a scalar or by another vector, but we cannot divide by a vector. The operation of division by a vector is not defined.
- 1 Vector Addition
- The addition, subtraction and multiplication of vectors are not the same as those of ordinary number. For example, if a long – distance runner runs 15 km east, 7km west and finally 4 km north. The net displacement is not 26 km. The computation has to take direction into account. Because of this, and for convenience's sake, we need to represent a vector by a line and an arrow. The length here denotes the magnitude of the vector while the arrow on it indicates direction.
- Algebra Vectors in One Dimension
- The addition of two vectors which are not inclined to each other but are parallel or anti parallel is as shown in the fig. 6 below
- Algebra of Vectors in Two Dimensions
- When the two vectors are at angle 90° or orthogonal, we complete the square or rectangle, then their addition called the resultant is the diagonal that is drawn from where the two tails of the vector meet.
- The Two methods of adding or subtracting vectors in two dimensions are Geometrical and analytical methods
- Geometrical Method:
- This method can be by construction of a Triangle or a Parallelogram.
- To be able to do this, two given vectors A and B that are at a particular angle to each other, will be represented by two adjacent sides of a triangle inclined to each other at the angle between the two vectors.

- The third side R of the triangle is the sum or resultant of the two vectors.
 - This is a triangle way of adding two vectors.
 - The parallelogram method is to draw the two vectors by two sides of the parallelogram and then, we complete the parallelogram.
 - The line drawn from where the two tails of the vectors joined called the diagonal is then the addition or the resultant of the two vectors in question.
- The triangle and parallelogram way of addition gives the same answer for the resultant or sum of the two vectors
- If we have more than two vectors A, B. and C their addition by construction is as indicated in Fig. 1.3

Multiplication of Vectors

As mentioned earlier, derived quantity is formed when two or more quantities are multiplied together. For example, multiplication of distance by force gives work and again the multiplication of distance by force also give moment or torque. Certainly, there must be difference in these two multiplication operations. There are generally the

- (a) Multiplication of vector by a scalar will produce a vector
- (b) Multiplication of a vector by a vector will produce a scalar (dot product)
- (c) Multiplication of a vector by a vector will produce a vector (cross product).

(a) Multiplication of Vector by a Scalar

A good example of this operation is that of the product of mass and acceleration to give force. A vector quantity the direction is in the direction of the vector being multiplied with a scalar.

(b) Multiplication of a vector by another Vector

This product is called dot product of two vectors. Because the resultant of this multiplication is a scalar, it is also called Scalar product.

Dot or Scalar product is defined as

θ = the angle between the two vectors.

If are defined as $\vec{A}$ and $\vec{B}$ respectively, then

$$= (\vec{A} \cdot \vec{B})$$

$$= (AB \cos \theta)$$

Because =

Example: A force $F = 5\hat{i} + 3\hat{j}$ acts on a body and changes its position from $P_1 (4,5)$ to $P_2 (-3,6)$ in a plane. Find the work done on that body if the force is newton and displacement is in meter. The displacement $s = P_1 - P_2 = \hat{i} (-3 - 4) + \hat{j} (6-5) = -7\hat{i} + \hat{j}$

$$\therefore W =$$

$$= (5 -7) + (3 = -32J)$$

) Multiplication of vector by vector (cross) product

This multiplication is defined as

Where $\hat{n}$ is a unit vector perpendicular to both $\vec{A}$ and $\vec{B}$. The magnitude of vector product is

The resultant of this multiplication is at right angle to both A and B. Vector product has the

property

If

Example: if $A = 2\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$ and $B = -\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}$. Find

$$= (-21 + 10) \mathbf{i}$$

$$= -11\mathbf{i} - 19\mathbf{j} - 7\mathbf{k}$$

// =

$$= 23.04$$

Coordinate Systems and Components of a Vector

Vectors are usually described in terms of their components in a coordinate system. In rectangular (Cartesian) xy plane, a point in a plane is described by a pair of coordinates (x, y). In like manner, a vector in a plane is described by a pair of its vector coordinates. The x -coordinate of vector A is called its x-component and the y coordinate is called the y component. The vector x component is a vector denoted by A_x . The y component is a vector denoted by A_y . In Cartesian system, the x and y components of a vector is the orthogonal projections of this vector onto the x- and y- axes respectively. In this way and using the parallelogram rule for vector addition, each vector in a Cartesian plane can be expressed as the vector sum of its vector components:

As shown in fig. 1.3 vector A is the diagonal of the rectangle where the x-component A_x is the side parallel to the x- axis and the y component A_y is the side parallel to the y-axis. Vector component A_x is orthogonal to vector component A_y

It is customary to denote the positive direction on the x-axis by the unit vector $\mathbf{i}$ and the positive direction on the y axis by the unit vector $\mathbf{j}$. Unit vectors of the axes, $\mathbf{i}$ and $\mathbf{j}$ define two orthogonal directions in the plane. From fig. 1.3, we can now write the x- and y- components of a vector in terms of the unit vector of the axes as

where A_x and A_y are numbers that define the vector components and are the scalar components of Vector A.

We obtain the component form of a vector

The scalar component of a vector by subtracting the origin point

coordinate from the end point coordinates i.e. $A_x = x_e - x_b$

and $A_y = y_e - y_b$

Example: The mouse of a computer is moved on the monitor from point (6.0cm, 1,6 cm) to an icon that is located at point (2.0cm, 4,5cm) What should be the displacement vector of the pointer?

Solution: $D_x = x_e - x_b = (2.0 - 6.0) \text{ cm} = -4.0\text{cm}$

$D_y = y_e - y_b = (4.5 - 1.6) \text{ cm} = +2.9\text{cm}$

The vector component form of the displacement vector is

$$D = D_x\mathbf{i} + D_y\mathbf{j} = (-4.0\text{cm})\mathbf{i} + (2.9\text{cm})\mathbf{j} = (-4.0\mathbf{i} + 2.9\mathbf{j}) \text{ cm}$$

We can find the magnitude of a vector when the scalar components are known. We use Pythagorean Theorem to find this magnitude as

$$A = \sqrt{A_x^2 + A_y^2} \quad A = \sqrt{A_x^2 + A_y^2}$$

This equation works if the scalar components are negatives. The direction of this magnitude is

obtained by finding the tangent function. So

A_y

$\tan \theta =$

A_x

This angle should be measured in the counterclockwise direction from the + x-axis to the vector.

From the example above the magnitude of the displacement vector $D = 4.9\text{cm}$ while its direction is -35.9° . This shows that vector D lies in the second quadrant, so its direction angle is

$$\theta_D = \theta + 180^\circ = -35.9^\circ + 180^\circ = 144.1^\circ$$

Vectors in Three Dimensions

An object in space is located by three coordinates (x, y, z) where coordinates x , and y specify locations in a plane, and coordinate z gives a vertical position above or below the plane. Three dimensional space has three orthogonal directions so we need not two but three unit vectors to define a three-dimensional coordinate system. In Cartesian coordinate system, the first two unit vectors are the unit vectors of x -axis i and the unit vector of the y -axis j . The third unit vector k is the direction of the z -axis. The order in which the unit vectors are labeled, which is the order in which the unit vectors appear is important because it defines the orientation of the coordinate system. The order x, y, z , which is equivalent to the order i, j, k , defines the standard right-handed coordinate system (positive orientation).

In three dimensional space, vector A has three vector components, the x -component $A_x = A_{xi}$, which is part of A along the x -axis, the y -component $A_y = A_{yj}$, which is part of A along the y -axis; and the z -component $A_z = A_{zk}$ which is part of A along the z -axis. A vector in three dimensional space is the vector sum of its three components

$$A = A_{xi} + A_{yj} + A_{zk}$$

The scalar component of the z -component is given by

$$A_z = z_e - z_b$$

If we know the beginning and the end coordinate $b(x_b, y_b, z_b)$ and $e(x_e, y_e, z_e)$ respectively since we have known that of A_x and A_y as discussed.

Magnitude A is obtained by generalizing to three dimensions

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2} \quad A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

- Mathematical Tools
- Introduction
- In physics, the knowledge of mathematical tool is necessary and important in order to treat and explain many concepts so that their meaning can be clearly explained. Such mathematical tools include derivatives (i.e. differentiation and integration) and trigonometric functions. These tools are only highlighted since their complete treatment is obtained in the relevant mathematics courses that we have offered or come across as we progress in our study of physics. So their treatment is beyond the scope of this course

1.1 Derivatives

When a graph of $y = 3x^2$ is plotted, the slope of this plotting when determine is not constant but

varies with the value of x . But the mathematician showed that it is possible to derive another function of x that will allow us to determine the slope at any value of x . The equation derived is called the derivative of the original equation. The process of deriving this equation is called

Differentiation and is represented as $\frac{dy}{dx}$. Mathematicians have shown that the first derivative of

the function $y = kx^n$

is

$\frac{dy}{dx}$

$= nkx^{n-1}$

1.1

$\frac{dy}{dx}$

$\frac{dy}{dx}$

For example, if $y = 6x^3$, then the derivative $= 18x^2$, while, $y = 3x^4 + 4x^3 + 2x^2 + 5x$, has the

$\frac{dy}{dx}$

$\frac{dy}{dx}$

derivative of $= 12x^3 + 12x^2 + 4x + 5$

$\frac{dy}{dx}$

Some common derivatives include

$\frac{dy}{dx}$

(i) $y = \cos x, \frac{dy}{dx} = -\sin x$ 1.2

$\frac{dy}{dx}$

$\frac{dy}{dx}$

(ii) $y = \sin x, \frac{dy}{dx} = \cos x$ 1.3

$\frac{dy}{dx}$

$\frac{dy}{dx}$

(iii) $y = A \cos bx, \frac{dy}{dx} = -Ab \sin bx$ 1.4

$\frac{dy}{dx}$

$\frac{dy}{dx}$

$y = A \sin bx, \frac{dy}{dx} = Ab \cos bx$ 1.5

$\frac{dy}{dx}$

Integration

This is the reverse of differentiation. Integration is denoted by the sign $\int$. Mathematicians have shown that

$\int w \, dx = \frac{w}{b+1} x + c$

1.6

$\int w \, dx = \frac{w}{b+1} x + c$

1.6

For example, $\int 12x^3 \, dx = 3x^4 + c$

A special integration is that of

1

$$\int \frac{1}{x} dx = \log a x + c \quad 17$$

The above is the general rule in which other common integration obey.

THANK YOU FOR LISTENING

Paper 4 of 7 — PHY101(Gravitation)

Level 100 • First Semester • Handout

3080

OSUN STATE UNIVERSITY

COLLEGE OF SCIENCE, ENGINEERING AND TECHNOLOGY

FACULTY OF BASIC AND APPLIED SCIENCES

DEPARTMENT OF PHYSICS

2024/2025 HARMATTAN SEMESTER

COURSE: GENERAL PHYSICS CODE: PHY 101 UNIT: 3

LECTURER: I.A. AKANNI (PhD)

3080 Course Contents

- Newton's law of gravitation
- Kepler's law of planetary motion
- Gravitational potential energy
- Escape Velocity
- Satellites motion and orbits

Newton's Law of Gravitation

3080 What is the Universal law of gravitation?

Newton's Law of Universal Gravitation states that every

particle attracts every other particle in the universe with force

F

F

directly proportional to the product of their masses (m_1 ; m_2)

and inversely proportional to the square of the distance (r^2)

between their centers. Proposed by Sir Isaac Newton in 1687

m_1 m_2

$F = G \frac{m_1 m_2}{r^2}$ (1)

G

G is Gravitational constant ($6.67 \times 10^{-11} \text{ N m}^2 / \text{kg}^2$); m_1 is the mass of the first body, m_2 is the mass of the second body, and r is the distance between the two bodies.

Law of Universal Gravitation

3080

Gravity varies depending on two factors: (i) The mass of an object doing the pulling (ii) The distance from the center of that object

Example:

1.The Moon is attracted to the Earth. The mass of the Earth is $6.0 \times 10^{24} \text{ kg}$ and the mass of the Moon is $7.4 \times 10^{22} \text{ kg}$. If the Earth and Moon are 345,000 km

apart, what is the gravitational force between them?

2. Calculate the gravitational force of attraction between the Earth and a 70 kg man standing at a sea level, a distance of 6.38×10^6 m from the earth's center.

Law of Universal Gravitation

3080

$\frac{1}{r^2}$

1. $\frac{1}{r^2} = \frac{1}{(6.38 \times 10^6)^2}$

$\frac{1}{r^2}$

$\frac{1}{(6.38 \times 10^6)^2}$

$\frac{1}{(6.38 \times 10^6)^2}$

$\frac{1}{(6.38 \times 10^6)^2} = 6.67 \times 10^{-11} \times$

$(3.45 \times 10^8)^2$

$\frac{1}{r^2} = 2.49 \times 10^{20} \text{ m}^{-2}$

$\frac{1}{r^2}$

2. $\frac{1}{r^2} = \frac{1}{(6.38 \times 10^6)^2}$

$\frac{1}{r^2}$

24

-11

$(5.98 \times 10^{24}) (70)$

$\frac{1}{r^2} = 6.673 \times 10^{-11} \times$

$(6.38 \times 10^6)^2$

$\frac{1}{r^2} = 685 \text{ m}^{-2}$

Gravitational Field

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A gravitational field is a region where a mass experiences a force due to gravitational attraction.

Note:

- A gravitational field is a region of space surrounding a body that has the property of mass. Any other body with mass will experience a force of attraction in this region.
- All bodies near the surface of Earth experience gravitation attraction exerted by Earth. The gravitational force or the weight acting on a mass m , in the gravitational field g , is given by:

$$F = mg \quad (2)$$

Gravitational Field

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- The direction of the gravitational field is always towards the center of the mass, causing the field.
- Gravitational forces are always attractive.
- Gravity has an infinite range, meaning it affects all objects in the universe

- There is a greater gravitational force around objects with a large mass (such as planets)
- There is a smaller gravitational force around objects with a small mass (almost negligible for atoms)

Gravitational Field

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- The gravitational field strength (gravitational acceleration) at a point is defined as the force per unit mass experienced by a test mass at that point.
- This can be written in equation form as:

$g = \frac{F}{m}$

$g = (3)$

$g =$

Where:

g is the gravitational field strength N kg^{-1}

F is the force due to gravity, or weight (N)

m is the mass of “test mass” in the field (kg)

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Gravitational Field

Interpretation:

- An object’s mass remains the same at all points in space. However, on planets such as Jupiter, the weight of an object will be greater than on a less massive planet such as Earth. This means the gravitational force would be so high that humans, for example, would not be able to stand up fully
- This is due to the increase in F , while m remains constant, ultimately leading to an increase in g felt

Exercise: Differentiate between mass and weight

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Gravitational Field

A body on Earth has a much

smaller force per unit mass than on

Jupiter

This means the body will have a much

greater weight on Jupiter than on

Earth

Earth Jupiter

$g_{\text{Earth}} = 9.81 \text{ N kg}^{-1}$ $g_{\text{Jupiter}} = 24.79 \text{ N kg}^{-1}$

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Gravitational Field

Mass:

- Mass is a measure of the amount of matter in an object. It is an intrinsic property of the object and doesn't change regardless of its location. Units: kg, g, etc.
- Has only magnitude and no direction (Scalar Quantity).
- The mass of an object is constant everywhere in the universe. Whether you are on Earth, on the Moon, or in space.
- Mass determines how much an object resists a change in its motion. This is called inertia. A more massive object will be harder to accelerate than a less massive one. This is the reason regardless of mass, all objects experiences $a = 9.8 \text{ m/s}^2$

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Gravitational Field

Weight:

- Weight is the force exerted on an object due to gravity. It depends on both the object's mass and the strength of the gravitational field it is in. Units: N
- Has both magnitude and direction (Vector Quantity).
- Weight depends on the gravitational pull of the object's location. For example, an object will weigh less on the Moon than on the Earth due to weak gravitation.
- It can be estimated using Eqn. 2.

Example 3: What is the weight of a 10 kg mass stone on (i) Earth (ii) Moon. Take

$$g_{\text{moon}} = 1.6 \text{ m/s}^2$$

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Gravitational Field

Solution:

Where:

From Eqn. 2; $F = \frac{GMm}{r^2}$ G = Newton's Gravitational Constant m (i) $m = 10 \text{ kg}$ $g = 9.81 \text{ m/s}^2 = 98.1 \text{ N}$ M = mass of the body causing the field (kg) r = distance from the mass where you are F (ii) $m = 10 \text{ kg}$ $g = 1.6 \text{ m/s}^2 = 16 \text{ N}$ calculating the field strength (m)

Since Eqn. 1 and 2 refers to the same force, equating them results in:

 $F = F$

$$\frac{GMm}{r^2} = mg \quad (4)$$

 G M

$$g = \frac{GM}{r^2} \quad (5)$$

 r

Gravitational Field

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From Eqn. 5 it can be concluded that the variables affecting the gravitational field

strength at the surface of a planet are:

1. The radius r (or diameter) of the planet
 2. The mass M (or density) of the planet
- Hence, objects with any mass m in that field will experience the same gravitational field strength

- The gravitational field strength g is inversely proportional to the square of the radial distance, $\propto \frac{1}{r^2}$

Kepler's law of Planetary motion

3080

- Kepler's Laws of Planetary Motion are three fundamental principles that describe the motion of planets around the Sun. They describe the motion and basic orbital mechanics of two-point source masses.
- In the Solar System, 8 planets orbit around one mass – the Sun. As the mass of the Sun is ~98% the mass of the entire Solar System, the gravitational forces on planets due to the mass of the other planets are negligible.
- Therefore, the scenario can be treated mathematically as two-point source masses when investigating the gravitational effects between the Sun and an orbiting planet.

Kepler's law of Planetary motion

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- These laws were formulated by the German astronomer Johannes Kepler in the early 17th century, based on the detailed observations of planetary positions made by the Danish astronomer Tycho Brahe.
- Therefore, the scenario can be treated mathematically as two-point source masses when investigating the gravitational effects between the Sun and an orbiting planet.

1. The Law of Orbits

Kepler's 1st law: Kepler's First Law of Planetary Motion is also known as the Law of Ellipses, states that all planets in the Solar System orbit the Sun in an elliptical

Kepler's law of Planetary motion

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1. The Law of Orbits

- An ellipse is an elongated or stretched circle. Unlike a circle with a constant radius, an ellipse has two axes.
- The semi-major axis: The longest radius of the ellipse, stretching from the center to the farthest point on the ellipse.
- The semi-minor axis: The shortest radius, stretching from the center to the closest point on the ellipse.
- Every ellipse has two focal points (or foci), which do not lie at the ellipse's center. In Kepler's First Law, the Sun is positioned at one of these foci.

Kepler's law of Planetary motion

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1. The Law of Orbits

- When a planet is closer to the Sun, it is at perihelion
- when it is farther away, it is at aphelion
- the general equation for an ellipse in a coordinate system (where the Sun is located at one of the foci) is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$a^2 - b^2 = c^2$$

$$c^2 = a^2 - b^2$$

$$c = \sqrt{a^2 - b^2}$$

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Kepler's law of Planetary motion

x and y are two points on the ellipse, a is the semi-major axis (half of the longest diameter of the ellipse), and b is the semi-minor axis (half of the shortest diameter of the ellipse).

Summary from First Law:

- The Sun is not at the center of the planetary orbit but at one of the two foci of an elliptical orbit.
- A planet's distance from the Sun varies as it orbits, meaning its speed changes as it moves closer to or farther from the Sun.

Kepler's law of Planetary motion

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2. Law of Equal Areas

Kepler's 2nd law: A line drawn from the center of the Sun to the center of an orbiting body (known as the radius vector) will sweep out equal areas in equal intervals of time.

Interpretation:

- The planet moves faster when it is closer to the Sun (at perihelion - early January) and slower when it is farther from the Sun (at aphelion - early July). A consequence of the gravitational force between the planet and the Sun, which is stronger when the planet is closer to the Sun.

Kepler's law of Planetary motion

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Interpretation:

- To satisfy Kepler's Second Law, the planet compensates for its faster speed near the Sun by traveling shorter distances over the same time interval.
- When it is farther from the Sun, it moves more slowly but covers a greater

distance in that same time interval. Ensuring that the area swept by the radius vector remains constant. Due to conservation of angular momentum.

- Mathematically expressed as;

$$\frac{dA}{dt}$$

$$= \frac{1}{2} \frac{d}{dt} (r^2 \frac{d\theta}{dt}) \quad (7)$$

$$\frac{dA}{dt}$$

$\frac{dA}{dt}$ is the differential area swept out by the radius vector, and $\frac{d\theta}{dt}$ is the differential

Kepler's law of Planetary motion

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Interpretation:

- time interval.

Aphelion

perihelion

Kepler's law of Planetary motion

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3. Harmonic Law

Kepler's 3rd law: The square of the period of a planet's orbit is directly proportional to the cube of the semi-major axis of its orbit. Mathematically;

$$T^2$$

$$= \frac{4\pi^2}{GM} a^3 \quad (8)$$

$$T^3$$

Where:

T is the orbital period of the planet, a is the semi-major axis of the planet's elliptical orbit. The constant depends on the mass of the Sun and the units used.

- The period T is the amount of time it takes a planet to complete one full orbit around the Sun. For Earth 1 year, meaning it takes 365.25 days for one full orbit.

Kepler's law of Planetary motion

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- Mercury takes 88 days, and Saturn takes 10,759 days
- The larger the orbital period, the further the planet is from the Sun
- The semi-major axis is the distance from the center of the ellipse (the average distance between the planet and the Sun). It is half of the longest diameter of the ellipse. The larger the semi-major axis, the larger the orbit.
- Since Eqn. 8 is constant for all planets orbiting the same star (the Sun, in our case), it can be inferred that for two planets;

$$T_1^2 \propto a_1^3$$

$$= 3 \quad (9)$$

$$T_2^2 \propto a_2^3$$

Kepler's law of Planetary motion

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E.g. for Earth and Mars: τ_1 is 1 year (Earth's orbital period); r_1 is 1 AU (Earth's semi-major axis); τ_2 is orbital period of Mars and r_2 is semi-major axis of Mars.

- Using the third law, astronomers can predict the orbital periods of planets, even those that haven't been observed yet, as long as the distance from the Sun is known.
- It can also be applied to moons orbiting planets or even artificial satellites orbiting Earth, as long as the mass of the central body (the Sun, Earth, etc.) is much larger than the mass of the orbiting body.

Kepler's law of Planetary motion

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Semi-Major axis

(r)

Ellipse centre

Minor axis

(r) Semi-minor

axis (r)

Major axis (r)

Kepler's law of Planetary motion

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From Eqn. 8, the proportionality constant is;

$$\tau^2 \propto r^3$$

$$= (10)$$

$$\tau^3 \propto r^2$$

The Law

of Periods

Example 4: Consider a planet in orbit around the Sun. If the planet's orbit is elliptical with a semi-major axis of 2 AU, what is the shape of the orbit based on the axes values? Assume the eccentricity is 0.3.

Kepler's law of Planetary motion

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Solution: Recall Kepler's First Law states that the orbit of every planet is an ellipse, with the Sun at one of the foci. An ellipse has two axes: a semi-major and a semi-minor axis. e describes how elongated the ellipse is.

e measures the amount of divergence from a perfect circle;

$$e^2$$

$$e = 1 - 2 \quad (11)$$

$$e$$

Where a is the semi-major axis, and b is the semi-minor axis

$$e = 2 \times 0.3$$

Hence, we obtain the semi-minor axis b

Kepler's law of Planetary motion

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$$e^2$$

$$0.3 = 1 - \frac{b^2}{a^2}$$

$$2$$

Square both sides,

$$e^2$$

$$0.09 = 1 - \frac{b^2}{a^2} \Rightarrow \frac{b^2}{a^2} = 0.91 \times 4 = 3.64$$

$$2$$

$$b \approx 1.91 \text{ AU}$$

Thus, the semi-minor axis b is approximately 1.91 AU. Since e is 0.3, the orbit is elliptical, but only slightly elongated.

Kepler's law of Planetary motion

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Example 5: Teaser question: A planet in orbit around the Sun moves from position A (at perihelion) to position B (at aphelion). If it takes 50 days to travel from A to B, how long will it take to travel from point B to the opposite side of the orbit (from aphelion to perihelion)?

Solution: ??

Kepler's law of Planetary motion

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Solution: Kepler's Second Law (Law of Equal Areas) states that a line joining the planet and the Sun sweeps out equal areas during equal time intervals. This means that the planet moves faster when it is closer to the Sun (at perihelion) and slower when it is farther away (at aphelion), but the areas swept by the radius vector are always equal. Since the planet sweeps out equal areas in equal time intervals, it will take the same time to travel from B to perihelion (opposite side of the orbit) as it did from A to B. Therefore, the time taken to travel from B to perihelion is also 50 days.

Kepler's law of Planetary motion

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Example 6: A planet is orbiting a star with a mass of $M = 2 \times 10^{30} \text{ kg}$. The semi-major axis of the planet's orbit is $a = 4 \text{ AU}$. What is the orbital period T of the planet?

Solution:

From Eqn. 10

$$T^2 = \frac{4\pi^2}{GM} a^3$$

$$3$$

□ □

Fitting the parameters; $G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$

Converting from AU to meters: $1 \text{ AU} = 1.496 \times 10^{11} \text{ m} \Rightarrow r = 5.984 \times 10^{11} \text{ m}$

Kepler's law of Planetary motion

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Sub. The values and make T the subject of the formula;

$r^3 = 4\pi^2 \frac{GM}{T^2}$

11^3

$5.984 \times 10^6 \cdot 6.674 \times 10^{-11} \times 2 \times 10^{30}$

$r^3 = 2.97 \times 10^{-19} \times 5.984 \times 10^{11}^3$

$r^3 \approx 6.35 \times 10^{15}$

$r = 2.52 \times 10^7 \text{ m}$

Convert seconds to years;

2.52×10^7

$r = \approx 0.8 \text{ years (or about 9.6 months)}$

$60 \times 60 \times 24 \times 365.25$

Kepler's law of Planetary motion

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Kepler's third law summary:

- The larger the orbit (greater semi-major axis r), the longer the planet's orbital period T . This relationship holds for any planet in the same star system.
- It was one of the major discoveries that laid the foundation for understanding the dynamics of planetary motion and helped in the later development of Newton's law of universal gravitation.

Gravitational Potential Energy

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- Gravitational potential energy is the energy possessed or acquired by an object due to a change in its position when it is present in a gravitational field. In simple terms, it can be said that gravitational potential energy is an energy that is related to gravitational force or to gravity.
- When a body of mass (m) is moved from infinity to a point inside the gravitational influence of a source mass (M) without accelerating it, the amount of work done in displacing it into the source field is stored in the form of potential energy.

Gravitational Potential Energy

3080

- Gravitational potential energy depends on:
 - ❖ Acceleration due to gravity (g)
 - ❖ The mass of the object measured in kilogram (m) and

❖ The height of the objects in meters (h)

• The gravitational potential (PEG) is given as:

$$PE = mgh \text{ ----- } -8$$

• Gravitational potential is the amount of work done in moving a unit test mass from infinity into the gravitational influence of source mass is known as gravitational potential.

Gravitational Potential Energy

3080

• The general expression for gravitational potential energy arises from the law of gravity and is equal to the work done against gravity to bring a mass to a given point in space. Because of the inverse square nature of the gravity force, the force approaches zero for large distances, and it makes sense to choose the zero of gravitational potential energy at an infinite distance away.

• The gravitational potential energy near a planet is then negative, since gravity does positive work as the mass approaches. This negative potential is indicative of a “bound state”; once a mass is near a large body, it is trapped until something can provide enough energy to allow it to escape. The general form of the gravitational potential energy of mass m is:

$$-m\phi$$

$$\phi = \text{-----} -9$$

Gravitational Potential Energy

3080 Note:

- The gravitational potential at a point is always negative, and V is maximum at infinity.
- The SI unit of gravitational potential is J/Kg.

Examples:

3. Calculate the gravitational potential energy of a body of mass 10 kg and is 25 m above the ground.

4. If the mass of the earth is 5.98×10^{24} kg and the mass of the sun is 1.99×10^{30} kg, and the earth is 160 million km away from the sun, calculate the gravitation potential energy of the earth.

5. A basketball weighing 2.2 kg falls off a building to the ground 50 m below. Calculate the gravitational potential energy of the ball when it arrives below.

Gravitational Potential Energy

3080

$$3. \quad PE = mgh$$

$$= 10 \times 9.8 \times 25$$

$$= 2450 \text{ J}$$

$$-2450 \text{ J}$$

$$4. \quad \phi =$$

$$(6.67 \times 10^{-11}) \times (5.98 \times 10^{24}) \times (1.99 \times 10^{30})$$

$$160 \times 10^9$$

$$= 4963 \times 10^{30} \text{ J}$$

$$5. \Delta = \Delta h$$

$$= 2 \times 12 \times 10 = 240 \text{ J}$$

The change in gravitational potential energy is equal to the work done by gravity. Therefore, Gravitational potential energy is equal to 240J

Escape Velocity

3080

- Escape Velocity is referred to as the minimum velocity needed by anybody or object to be projected to overcome the gravitational pull of the planet earth. In other words, the minimum velocity that one requires to escape the gravitational field is escape velocity.
- It refers to the initial velocity a particle must have at the surface of a planet in order to leave the planet forever, and never be pulled back by the planet's gravity.
- If a particle leaves the surface of a planet with an initial velocity equal to the escape velocity, then the body will move more and more slowly as the particle moves farther from the planet, finally reaching a velocity of zero at $r=\infty$. (We assume only the particle and the planet are present, and ignore all other bodies.)

Escape Velocity

3080

- Escape velocity formula is given as:

$$2\Delta$$

$$\Delta = \frac{GM}{R} - \frac{GM}{\infty} = \frac{GM}{R}$$

$$\Delta$$

Where,

V is the escape velocity

G is the gravitational constant is $6.67408 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

M is the mass of the planet

R is the radius from the center of gravity

- It is expressed in m/s and the escape velocity of earth is 11,200 m/s.

Escape Velocity

3080

- An alternative expression for the escape velocity particularly useful at the surface on the body is

$$v_{\text{esc}} = \sqrt{2gR}$$

Where g is the acceleration due to the gravity of earth.

Hence, Escape velocity is also given by : $v_{\text{esc}} = \sqrt{2gR}$

Examples

6. Determine the escape velocity of the Jupiter if its radius is 7149 Km and mass is $1.898 \times 10^{27} \text{ kg}$

7. Determine the escape velocity of the moon if Mass is $7.35 \times 10^{22} \text{ kg}$ and the radius is $1.5 \times 10^6 \text{ m}$

Escape Velocity

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8. Suppose the radius of Earth is 6.38×10^6 m and the mass of the planet earth is 5.98×10^{24} kg. Find out the escape velocity from planet earth.

9. In order to leave the moon, the Apollo astronauts had to take off in the lunar mobile and reach the escape velocity of the moon. The radius of the moon is 1.74×10^6 m, and the mass of the moon is 7.35×10^{22} kg. Calculate the velocity which the Apollo astronauts have to reach in order to leave the moon?

Escape Velocity

200

3080 6. 0000 =

0

$$2 \times (6.67408 \times 10^{-11}) \times (1.898 \times 10^{27})$$

$$7149 \times 10^3$$

$$= 188251 \text{ m/s}$$

200

7. 0000 =

0

$$2 \times (6.67408 \times 10^{-11}) \times (7.35 \times 10^{22})$$

$$1.5 \times 10^6$$

$$= 2557 \text{ m/s}$$

Escape Velocity

200

8. 0000 =

3080

0

$$2 \times (6.67408 \times 10^{-11}) \times (5.98 \times 10^{24})$$

$$6.38 \times 10^6$$

$$= 11185 \text{ m/s}$$

200

9. 0000 =

0

$$2 \times (6.67408 \times 10^{-11}) \times (7.35 \times 10^{22})$$

$$1.74 \times 10^6$$

$$= 2374 \text{ m/s}$$

Satellite Motion and Orbits

3080

- The satellites revolve around the earth, making a circular or elliptical path called the orbit.

- Any heavenly or human-made body that revolves around a central body, such as the Earth, Sun, or other bodies in the solar system, is known to be a satellite. However, these satellites revolve around and make either circular or elliptical orbits when it comes to Earth. The earth, in this case, is said to be the central body.
- Satellites are objects or bodies that revolve around the heavier body in space.

Satellite Motion and Orbits

3080 There are two types of satellites:

- Natural Satellites: Satellites naturally present in space, such as comets and the Moon.
- Artificial Satellites: Satellites made by humans for various purposes such as weather forecasting, communication, etc.
- Every planet has at least one of its natural satellites. The earth has only one, the Moon, while other planets such as Jupiter are known to have 53 natural satellites (moons).
- The Soviet Union, now Russia, launched the first artificial satellite into space on Oct 4, 1957, named Sputnik 1. The launch of Sputnik 1 triggered what was called the Space Race.

Motion of Satellite around the earth

- 3080 • The satellite orbiting the earth has a velocity and acceleration component similar to an object moving on a circular path. While revolving around the earth, satellites make a circular path.
- The tangent to this circular path gives the direction of the satellite's velocity, whereas acceleration is towards the centre of the circle.
 - Various forces act between the Earth and the satellite, making them orbit on a circular path. The net force that acts in the inward direction of the circle makes the satellite accelerate around the Earth.

Motion of Satellite around the earth

- The gravitational force that acts between the Earth and the satellite
- 3080

and centripetal force are two main forces causing the motion of satellites around Earth.

- If there is no centripetal force, satellites will keep moving at the same speed and direction. In such a case, the satellite will move on a straight path, and there will be no circular motion.
- However, some satellites move along an elliptical orbit. In that case, unlike the circular motion, the central body is positioned at

foci of the ellipse.

- The velocity component of the satellite acts along the tangent of the ellipse. At the same time, the acceleration acts towards the focus.

Motion of Satellite around the earth

- Based on Newton's Second Law of Motion, the direction of the net force
- 3080

will be towards the focus.

- This is because, along with centripetal force and gravitational force, a force component acts either in the same or opposite direction.
- This component of force changes the speed of the satellite. Hence, satellites with elliptical motion do not have a constant speed.

Orbital Velocity of Satellite

- The velocity with which satellites move around the earth is known as the orbital velocity of satellites.
- Orbital velocity is also known as critical velocity. The minimum velocity required to make a satellite move around the earth with stability.

Paper 5 of 7 — PHY 101 LECTURE 3

Level 100 • First Semester • Handout

3080

OSUN STATE UNIVERSITY

COLLEGE OF SCIENCE, ENGINEERING AND TECHNOLOGY

FACULTY OF BASIC AND APPLIED SCIENCES

DEPARTMENT OF PHYSICS

2023/2024 HARMATTAN SEMESTER

COURSE CODE: PHY 101 COURSE: GENERAL PHYSICS 1 UNIT: 3

LECTURERS: DR. J.T. ADELEKE

3080 WORK

□ Work W is the energy transferred to or from an object by means of a force acting on the object. Energy transferred to the object is positive work, and energy transferred from the object is negative work. It is a scalar quantity

□ The work done by a force can be defined as the product of the magnitude of the displacement and the component of the force in the direction of the displacement. The unit of work in S.I. is the Joule (J). $1 \text{ J} = 1 \text{ N.m}$ other unit of work is foot-pound (ft.lb), although old use.

3080

Mathematical expression of Work done

F

□□

□

□ □

f

Fig. 1 □ □□

Work can be expressed mathematically as Fig. 2 i

$$\square = \square \square \square \square \dots (1)$$

$$\square = \square \cdot \square \square \dots (2)$$

Equation (2) is especially useful for calculating the work when □□ □□ □□ are given in unit vector notations.

If a force displaces the particle through a distance dr , then the work done dW by the forced is

□ □□□

$$\square \square = \square \dots (3)$$

The total work done by the force in moving the particle from initial point I to final point f will be

□ □

$$\vec{F} = F \cos \theta \cdot \vec{s} = F s \cos \theta \dots (4)$$

WORK-KINETIC ENERGY THEOREM

3080

The work-kinetic energy theorem states that: “when a body is acted upon by a force or resultant force, the work done by the force is equal to the change in kinetic energy of the body”.

□ □

Proof: $\vec{F} = F \cos \theta \cdot \vec{s} = F s \cos \theta$

Force $\vec{F}$ is in the direction of displacement $\vec{s}$,

$$\theta : \theta = 0$$

□ □

$$W = F s \cos \theta = F s$$

□ □

□ □

□ □ □ □

$$W = \Delta K = K_2 - K_1$$

□ □ □ □ □ □

□ □ □

□ □

$$W = \Delta K = K_2 - K_1 = \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2$$

□ □ □ □

□

$$W = \Delta K = K_2 - K_1$$

$$W = \Delta K = K_2 - K_1$$

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$$W = \Delta K = K_2 - K_1$$

$$W = \Delta K = K_2 - K_1 \dots (5)$$

$$W = \Delta K = K_2 - K_1$$

$$W = \Delta K = K_2 - K_1 \dots (6)$$

3080

POWER

Power is the time rate at which work is done. If an amount of work ΔW is done in a small interval of time Δt . It is the scalar product of $\frac{\Delta W}{\Delta t}$ □.

□ The S.I. unit of

power is J/S, which 1 Watt. (1 W = 1 J/S). Power P is expressed as

$$P = \frac{\Delta W}{\Delta t}$$

$$P = \lim_{\Delta t \rightarrow 0} \frac{\Delta W}{\Delta t} \dots (7)$$

$$\Delta t \rightarrow 0 \Delta W \dots$$

$$\Delta W$$

$$W = \Delta U = \Delta K.$$

$$= \Delta U + \Delta K \quad (8)$$

$$\Delta U = \Delta K$$

Recall that instantaneous velocity

$$v =$$

$$V =$$

$$v =$$

If $\vec{F}$ acts at an angle of θ to v , then

$$F_{\text{parallel}} = F \cos \theta \dots\dots\dots$$

$$P = Fv \quad (9)$$

3080

POTENTIAL ENERGY (U), POSITIVE, NEGATIVE AND ZERO WORK

☐ Potential energy U is associated with system of bodies or objects under the action of forces.

☐ Assuming a woman who carries a load on her head is static. She does no work so long she makes no displacement, although she may be sweating and tired!

☐ This is because the component of the force on her is vertical, i.e. $\theta = 0$ and $\cos \theta = 1$.

☐ The work done on the load by her neck/head is negative. You can remember Newton's third law of motion supporting this.

$$\Delta U = -W$$

Remember that

$$W =$$

$$W = \int \vec{F} \cdot d\vec{r}$$

$$W =$$

$$W =$$

$$\therefore \Delta U = -W$$

$$W =$$

3080 POTENTIAL ENERGY (U), POSITIVE, NEGATIVE AND ZERO WORK

If gravitational, it is the potential energy associated with objects under the influence of gravitational force, i.e a system of Earth and nearby particle.

$$W =$$

$$\Delta U = -W$$

$$W =$$

$$W =$$

$$= - \int \vec{F} \cdot d\vec{r}$$

$$W =$$

$$W =$$

$$= \int_{r_1}^{r_2} \frac{GMm}{r^2} dr$$

$$W =$$

$$\Delta U = \int_{r_1}^{r_2} \frac{GMm}{r^2} dr$$

Thus, taking the starting point to be zero, then

$$\Delta U = \frac{GMm}{r}$$

3080 ELASTIC POTENTIAL ENERGY

- ☐ Elastic potential energy is the energy stored up in elastic materials.
- ☐ It is associated with the state of the object whether compressed or stressed.
- ☐ For a spring which exerts a force $F = -kx$

$W =$

$$\Delta W = - \int_{x_1}^{x_2} F dx$$

$W =$

$W =$

$$= - \int_{x_1}^{x_2} (-kx) dx$$

$W =$

$W =$

$$= \frac{1}{2} k (x_2^2 - x_1^2)$$

$W =$

$W =$

$$= \frac{1}{2} k x^2$$

$$= \frac{1}{2} k x^2$$

$$= \frac{1}{2} k x^2$$

$$= \frac{1}{2} k x^2$$

$$= \frac{1}{2} k x^2 - \frac{1}{2} k x^2$$

3080 WORKED EXAMPLES

Example 1:

A boy pushes a box with a force of 120 N at an angle of 30° to the horizontal, through a distance of 10 m.

how much work is done by the boy?

Solution: $W = F \cos \theta \cdot d$

$$W = (120 \text{ N})(10 \text{ m}) \cos 30^\circ$$

$$W = 1.039 \times 10^3 \text{ J}$$

Example 2:

A girl pulls her luggage with a constant force $F = 80 \text{ N}$ at an angle of $\theta = 30^\circ$ through a displacement of $d =$

10 m at an angle of $\theta = 120^\circ$. How much work is done by the girl on her luggage?

Solution:

$$W = F \cos \theta \cdot d + F \cos \theta \cdot d + F \cos \theta \cdot d$$

Recall that $P = F \cos \theta$.

$$= (80 \text{ N})(10 \text{ m}) + (-40 \text{ N})(12 \text{ m})$$

$$= 3.2 \times 10^3 \text{ J}$$

3080 CLASS WORK

Class work 1:

A student representative, drags a public address (PA) system weighing 7000 N through a distance of 20 m to the PHY 101 lecture theater. The student rep exerts a force of 3000 N at an angle of 40° to the horizontal. If a 2000 N

frictional force opposes the motion of the public address system, find the work done by each force acting on the PA and the total work done by all the forces.

Class work 2:

A particle is acted upon by the forces

$$\vec{F}_1 = 5\hat{i} - 10\hat{j} + 15\hat{k}$$

$$\vec{F}_2 = 10\hat{i} - 25\hat{j} - 20\hat{k}, \text{ } \vec{F}_3 =$$

$$15\hat{i} - 20\hat{j} + 10\hat{k}$$

Find the force needed to keep the particles in equilibrium.

3080

SOLUTION TO CLASS WORKS

Class work 1 solution

Work done by the weight is zero, Work done by normal force is also zero

It remains work done by the force exerted by the student rep and the work done by the friction.

Thus, the total active force acting on the PA is

$$\vec{F} = \vec{F}_{\text{student}} + (-\text{Frictional force})$$

$$\vec{F} = 3000 \text{ N} \cos 400 - 2000 \text{ N}$$

$$\vec{F} = 298.13 \text{ N}$$

$$\text{The total work done is } W = \vec{F} \cdot \vec{d} = (298.13 \text{ N})(20 \text{ m}) = 5962.6 \text{ J}$$

Class work 2 solution

The resultant force is R

$$R = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 5\hat{i} - 10\hat{j} + 15\hat{k} + 10\hat{i} + 25\hat{j} - 20\hat{k} + (15\hat{i} -$$

$$20\hat{j} + 5\hat{k})$$

$$= 30\hat{i} - 5\hat{j} + 5\hat{k}$$

The force to keep this in equilibrium

$$= -\vec{F} = -30\hat{i} + 5\hat{j} - 5\hat{k}$$

3080

Nonconservative Forces

□ A nonconservative force is one for which work depends on the path taken.

□ Friction is a good example of a nonconservative force.

□ An important characteristic is that the work done by a nonconservative force adds or removes mechanical energy from a system.

□ Friction, for example, creates thermal energy that dissipates, removing energy from the system.

The amount of the happy face erased depends on the path taken by the eraser between points A and B, as does the

work done against friction. Less work is done and less of the face is erased for the path in (a) than for the path in (b).

The force here is friction, and most of the work goes into thermal energy that subsequently leaves the system (the

happy face plus the eraser). The energy expended cannot be fully recovered.

3080 How Nonconservative Forces Affect Mechanical Energy

- ☐ Mechanical energy may not be conserved when nonconservative forces act.
- ☐ For example, when a car is brought to a stop by friction on level ground, it loses kinetic energy, which is dissipated as thermal energy, reducing its mechanical energy.

Comparison of the effects of conservative and nonconservative forces on the mechanical energy of a system. (a) A

system with only conservative forces. When a rock is dropped onto a spring, its mechanical energy remains constant

(neglecting air resistance) because the force in the spring is conservative. The spring can propel the rock back to its

original height, where it once again has only potential energy due to gravity. (b) A system with nonconservative forces.

When the same rock is dropped onto the ground, it is stopped by nonconservative forces that dissipate its mechanical

energy as thermal energy, sound, and surface distortion. The rock has lost mechanical energy.

3080

Linear Momentum and Force

Linear momentum is defined as the product of a system's mass multiplied by its velocity. In symbols, linear momentum is expressed as $p = mv$.

The SI unit for momentum is $\text{kg} \cdot \text{m/s}$

Example: (a) Calculate the momentum of a 110-kg football player running at 8.00 m/s. (b) Compare the player's

momentum with the momentum of a hardthrown 0.410-kg football that has a speed of 25.0 m/s.

Solution for (a)

To determine the momentum of the player, substitute the known values for the player's mass and speed into the

equation. $p = mv$

$$p_{\text{player}} = (110 \text{ kg})(8.00 \text{ m/s}) = 880 \text{ kg} \cdot \text{m/s}$$

Solution for (b)

To determine the momentum of the ball, substitute the known values for the ball's mass and speed into the equation.

$$p = mv$$

$$p_{\text{ball}} = (0.410 \text{ kg})(25.0 \text{ m/s}) = 10.3 \text{ kg} \cdot \text{m/s}$$

Discussion

Although the ball has greater velocity, the player has a much greater mass. Thus the momentum of the player is much

greater than the momentum of the football, as you might guess. As a result, the player's motion is only slightly affected if

he catches the ball. We shall quantify what happens in such collisions in terms of momentum in later sections.

3080

Momentum and Newton's Second Law

Newton stated his second law of motion in terms of momentum: The net external force equals the change in

momentum of a system divided by the time over which it changes. Using symbols, this law is

$$F_{\text{net}} = \Delta p / \Delta t,$$

where F_{net} is the net external force, Δp is the change in momentum, and Δt is the change in time.

Linear Momentum and Force

Change in momentum Δp is given by $\Delta p = \Delta(mv)$.

If the mass of the system is constant, then $\Delta(mv) = m\Delta v$.

So that for constant mass, Newton's second law of motion becomes $F_{\text{net}} = \Delta p / \Delta t = m\Delta v / \Delta t$.

Because $\Delta v / \Delta t = a$, $\therefore F_{\text{net}} = ma$

3080 WORKED EXAMPLES

Example 1: During the in the US Open at USTA Billie Jean King National Tennis Center on September 8, 2022, Ons

Jabeur of Tunisia hit the fastest recorded serve in a premier women's match, reaching a speed of 58 m/s (209 km/h).

What is the average force exerted on the 0.057-kg tennis ball by Ons Jabeur's racquet, assuming that the ball's speed

just after impact is 58 m/s, that the initial horizontal component of the velocity before impact is negligible, and that

the ball remained in contact with the racquet for 5.0 ms (milliseconds)?

Solution

When mass is constant, the change in momentum is given by $\Delta p = m\Delta v = m(v_f - v_i)$.

To determine the change in momentum, substitute the values for the initial and final velocities into the equation,

$$\Delta p = m(v_f - v_i)$$

$$= (0.057 \text{ kg})(58 \text{ m/s} - 0 \text{ m/s})$$

$$= 3.306 \text{ kg} \cdot \text{m/s} \approx 3.3 \text{ kg} \cdot \text{m/s}$$

Now the magnitude of the net external force can be determined by using $F_{\text{net}} = \Delta p / \Delta t$:

$$F_{\text{net}} = \Delta p / \Delta t = 3.306 \text{ (kg} \cdot \text{m/s)} / 5.0 \times 10^{-3} \text{ s}$$

$$= 661 \text{ N} \approx 660 \text{ N}$$

3080

Conservation of Momentum

Consider what happens if the masses of two colliding objects are more similar than the masses of a football player and Earth—for example, one car bumping into another

A car of mass m_1 moving with a velocity of v_1 bumps into another car of mass m_2 and velocity v_2 that it is following. As a

result, the first car slows down to a velocity of v'_1 and the second speeds up to a velocity of v'_2 .

The momentum of each

car is changed, but the total momentum p_{tot} of the two cars is the same before and after the collision (if you assume

friction is negligible).

3080

Change in Momentum

$$\Delta p_1 = F_1 \Delta t,$$

$$\Delta p_2 = F_2 \Delta t$$

where F_2 is the force on car 2 due to car 1, and we assume the duration of the collision Δt is the same

for both cars.

We know from Newton's third law that $F_2 = -F_1$,

$$\text{and so } \Delta p_2 = -F_1 \Delta t = -\Delta p_1$$

Thus, the changes in momentum are equal and opposite, and $\Delta p_1 + \Delta p_2 = 0$.

Because the changes in momentum add to zero, the total momentum of the two-car system is constant. That is,

$$p_1 + p_2 = \text{constant},$$

$$p_1 + p_2 = p'_1 + p'_2,$$

where p'_1 and p'_2 are the momenta of cars 1 and 2 after the collision.

In equation form, the conservation of momentum principle for an isolated system is written

$$p_{\text{tot}} = \text{constant},$$

$$\text{Or } p_{\text{tot}} = p'_{\text{tot}},$$

where p_{tot} is the total momentum and p'_{tot} is the total momentum some time later. An isolated system

is defined to be one for which the net external force is zero $F_{\text{net}} = 0$.

Elastic Collision

3080

An elastic collision is one that conserves internal kinetic energy.

CONSERVATION OF MOMENTUM

When a collision occurs in an isolated system, the total momentum of the system doesn't change with the passage of

time. Instead, it remains constant both in magnitude and in direction. The momenta of the individual objects in the

system may change, but the vector sum of all the momenta will not change. The total momentum, therefore, is said to

be conserved.

When no net external force acts on a system, the total momentum of the system remains constant in time.

3080

WORKED EXAMPLE

A 0.150-kg baseball, thrown with a speed of 40.0 m/s, is hit straight back at the pitcher with a speed of 50.0 m/s.

(a) What is the impulse delivered by the bat to the baseball?

(b) Find the magnitude of the average force exerted by the bat on the ball if the two are in contact for 2.00 $\times 10^{-3}$ s.

Solution (a) Find the impulse delivered to the car.

Calculate the initial and final momenta of the car:

(b) Find the average force exerted on the car.

Apply Equation 6.6, the impulse-momentum theorem:

The impulse is just the difference

between the final and initial momenta:

3080

CLASS WORK

1. An archer stands at rest on frictionless ice and fires a 0.500-kg arrow horizontally at 50.0 m/s.

(See Fig.6) The combined mass of the archer and bow is 60.0 kg. With what velocity does the archer move across the ice after firing the arrow? (Fig.6)

2. A 70.0-kg man and a 55.0-kg woman on ice skates stand facing each other. If the woman pushes the man backwards so that his final speed is 1.50 m/s, at what speed does she recoil?

3. A car and a large truck traveling at the same speed collide head-on and stick together. Which vehicle experiences the larger change in the magnitude of its momentum? (a) the car

(b) the truck (c) the change in the magnitude of momentum is FIG. 7 (a) Before and (b) after a perfectly

the same for both (d) impossible to determine inelastic head-on collision between two objects.

CENTER OF MASS AND SYSTEMS OF PARTICLES

☐ The center of mass of a body is defined as a point where the entire mass of the body appears to be concentrated.

☐ Therefore, this point can represent the entire body.

■ For bodies of regular shape and uniform mass distribution, the center of mass is at the geometric center of the body. Some examples are tabulated below:

Bodies of regular shape and Geometric center of the body
uniform mass distribution

circle and sphere Centers of the circle and sphere

square and rectangle Point of intersection of their diagonals

cube and cuboid Point of intersection of their body
diagonals

Center of Mass for Distributed Point Masses

■ A point mass is a hypothetical point particle which has nonzero mass and no size or shape.

- To find the center of mass for a collection of n point masses, say, $m_1, m_2, m_3 \dots m_n$ we have to first choose an origin and an appropriate coordinate system as shown in Fig. C1
- Let, $x_1, x_2, x_3 \dots x_n$ be the X-coordinates of the positions of these point masses in the X direction from the Fig. C1: center of mass for a origin. collection of n point masses

Center of Mass for Distributed Point Masses Cont.

The equation for the X coordinate of the center of mass is,

Hence, the position of center of mass of these point masses in a Cartesian coordinate system is (x_{CM}, y_{CM}, z_{CM}) .

Position of Center of Mass

Generally, the position of center of mass can be written in a vector form as,

Center of Mass of Two Point Masses

The center of mass of two point masses m_1 and m_2 , which are at positions x_1 and x_2 respectively on the X-axis.

- (i) When the masses are on positive X-axis:
- (ii) When the origin coincides with any one of the masses:
- (iii) When the origin coincides with the center of mass itself:

The equation is principle of moments.

Ex 1. A uniform disc of mass 100g has a diameter of 10 cm. Calculate the total energy of the disc when rolling along a horizontal table with a velocity of 20 cm s^{-1} . (take the surface of table as reference)

Mass of the disc $m = 100 \text{ g} = 0.1 \text{ kg}$.

Diameter of the disc $d = 10 \text{ cm}$

Radius of the disc $r = 5 \text{ cm} = 0.05 \text{ m}$

Rolling with a velocity $v = 20 \text{ cm s}^{-1} = 0.20 \text{ m s}^{-1}$

Total energy of the disc $E_{\text{Tot}} = ?$

$E_{\text{Tot}} = \text{Translational K.E.} + \text{rotational K.E.}$

Moment of inertia (M.I) of the disc about

its own axis

$\therefore I = 1.028$ units

Ex 2. A particle of mass 5 units is moving with a uniform speed of $v = 3\sqrt{2}$ units in the XOY plane along the line $y = x + 4$. Find the magnitude of angular momentum.

Ex 3. A fly wheel rotates with a uniform angular acceleration. If its angular velocity increases from 20π rad/s to 40π rad/s in 10 seconds. Find the number of rotations in that period.

Initial Angular Velocity $\omega_0 = 20\pi$ rad s⁻¹

Final Angular velocity $\omega = 40\pi$ rad s⁻¹

time taken, $t = 10$ s

No. of rotations / s = ?

Solved Problems

Ex.1. Two point masses 3 kg and 5 kg are at 4 m and 8 m from the origin on X-axis. Locate the position of center of mass of the two point masses (i) from the origin and (ii) from 3 kg mass.

Solution

Let us take, $m_1 = 3$ kg and $m_2 = 5$ kg

(i) To find center of mass from the origin:

The point masses are at positions, $x_1 = 4$ m, $x_2 = 8$ m from the origin along X axis.

CLASS WORK 1

No 1

A. 0 B. 6 C. 8 D. 2 E. 1

SOLUTION TO CLASS WORK 1

CLASS WORK 2

A. 5 m B. $10/3$ m C. $20/3$ m D. 10 E. 2

SOLUTION TO CLASS WORK 2

Paper 6 of 7 — PHY 101 GENERAL PHYSICS 1

Level 100 • First Semester • Handout

3080

OSUN STATE UNIVERSITY

COLLEGE OF SCIENCE, ENGINEERING AND TECHNOLOGY

FACULTY OF BASIC AND APPLIED SCIENCES

DEPARTMENT OF PHYSICS

2023/2024 HARMATTAN SEMESTER

COURSE CODE: PHY 101 COURSE: GENERAL PHYSICS 1 UNIT: 3

LECTURERS: DR. J.T. ADELEKE

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Course Contents

- Conservation principles in physics, conservative forces,
- Conservation of linear momentum
- Kinetic energy and work, Potential energy,
- System of Particles, Centre of mass.
- Rotational motion. Torque,
- Vector product, moment,
- Rotation of Coordinate axes and angular momentum,
- Polar coordinates,
- Conservation of angular momentum,

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Further reading/References

- College Physics, 7th Edition - Serway, Raymond A.; Faughn, Jerry S.; Vuille, Chris; Bennett, Charles A.
- Fundamentals of Physics, Tenth Edition - David Halliday, Robert Resnick -

https://elearn.daffodilvarsity.edu.bd/pluginfile.php/987150/mod_label/intro/fundamentals-of-physics-textbook.pdf

Conservation principles in physics - Energy

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- In general, a conservation law is a statement that a certain quantity does not change over time. principle that states that a certain physical property (i.e., a measurable quantity) does not change in the course of time within an isolated physical system.
- The principle governs energy, momentum, angular momentum, mass, and electric charge. In particle physics, other conservation laws apply to properties of subatomic particles that are invariant during interactions.
- Conservation of energy implies that energy can be neither created nor destroyed,

although it can be changed from one form (mechanical, kinetic, chemical, etc.) into another. For example, a falling body has a constant amount of energy, but the form of the energy changes from potential to kinetic.

- when a system is isolated from its surroundings, the energy of the system is always conserved

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Conservation principles - Mass

Law of Conservation of Mass

- This law states that mass can neither be destroyed nor be created in a chemical reaction. i.e., processes that change the physical or chemical properties of substances within an isolated system (such as conversion of a liquid to a gas) leave the total mass unchanged
- Atoms rearrange themselves in different fashions among different molecules, in a chemical reaction, hence the mass remains constant.
- Since the mass remains constant, the total reactant mass and the total product mass remain the same. Chemical reactions can be both exothermic and endothermic.
- Both the laws of conservation of mass and conservation of energy can be combined into one law, the conservation of mass-energy.

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Conservation principles - linear momentum

- Conservation of linear momentum expresses the fact that a body or system of bodies in motion retains its total momentum, the product of mass and velocity, unless an external force is applied to it.
- In an isolated system (such as the universe), there are no external forces, so momentum is always conserved. Because momentum is conserved, its components in any direction will also be conserved.
- Applications are in the solution of collision problems.
- The operation of rockets exemplifies the conservation of momentum: the increased forward momentum of the rocket is equal but opposite in sign to the momentum of the ejected exhaust gases.

3080 Conservation principles - angular momentum

- Conservation of angular momentum of rotating bodies is analogous to the conservation of linear momentum.
- Angular momentum is a vector quantity whose conservation expresses the law that a body or system that is rotating continues to rotate at the same rate unless a twisting force, called a torque, is applied to it.
- The angular momentum of each bit of matter consists of the product of its mass, its distance from the axis of rotation, and the

component of its velocity perpendicular to the line from the axis.

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Conservation principles - charge

- Conservation of charge states that the total amount of electric charge in a system does not change with time.
- At a subatomic level, charged particles can be created, but always in pairs with equal positive and negative charge so that the total amount of charge always remains constant.
- Other conservation laws apply to certain properties of nuclear particles, such as baryon number, lepton number, and strangeness.

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Conservation principles - Derivation

Mechanical Energy = Potential energy + kinetic energy

Potential energy is given by $U = mgh$,

Kinetic energy is given by $K = \frac{1}{2}mv^2$,

Consider a freely falling body,

The force exerted on the body is $F = mg$

Now, if the body falling is from height H , the potential energy is $U_1 = mgH$

potential energy at time t and height h is, $U_2 = mgh$

change in potential energy $(\Delta U) = U_1 - U_2$

$F = ma$ and a (acceleration due to gravity) $= v \cdot dv/dx$

$F = m \cdot v \cdot dv/dx$

Now on integration on both sides,

$\int F \cdot dx = \int v^2 v_1 m \cdot v dv$

Here, $F = mg$ and v_1 and v_2 is 0 and v respectively,

$\int mg \cdot dx = \int v_0 m \cdot v dv$

$mgx = \frac{1}{2}mv^2$

Conservative Force

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- A conservative force is a force done in moving a particle from one point to another, such that the force is independent of the path taken by the particle. It depends only on the initial and final position of the particle. That is the work done is only dependent upon the starting and ending positions, not the path taken to get there.
- Examples are; the magnetic force, electrostatic force, gravitational force, etc.
- When the work done by a force is dependent on the

path taken, this force is a non-conservative force. Air Fig. 1: Gravitational force acting resistance is an example of a non-conservative force. on a particle

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Properties of Conservative Forces

The following are the properties of a conservative force:

- When the force depends only on the initial and final position irrespective of the path taken.
- In any closed path, the work done by a conservative force is zero.
- The work done by a conservative is reversible.

Examples of Conservative Force: 1. Weight of an Object, 2. Stretching an Elastic Band, 3. Poles of a Magnet, 4. Charged Comb, 5. Objects Falling on the Ground, 6. Spring Action

Properties of Non-Conservative Forces

The properties are given below:

- It is path-dependent therefore it also depends on the initial and final position.
- In any closed path, the total work done by a non-conservative force is not zero.
- The work done by a non-conservative is irreversible.

Differences between Conservative and Nonconservative Forces

The work done by conservative force The work done by nonconservative force

The work done between two The work done between two
points by any conservative points by any conservative
force force

- a) is reversible. a) is NOT reversible.
- b) is independent of the path b) is DEPENDENT of the path
between the two points. between the two points.
- c) is zero if the starting and c) is NOT zero if the starting
ending points are the and ending points are the
same. same.

Solved Problems

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Example 1

So the work done by the push force
is:

The friction force vector points in the opposite
direction compared to the motion of the chair, so it
does negative work on the chair:

Taking the sum of these, we get the total work done
by the non-conservative forces on the chair:

Solved Problems

Example 2: Consider an object of mass 2 kg moved by an external force 20 N in a surface having

coefficient of kinetic friction 0.9 to a distance 10 m. What is the work done by the external force and

kinetic friction ? Comment on the result. (Assume $g = 10 \text{ ms}^{-2}$)

Solution

$m = 2 \text{ kg}$, $d = 10 \text{ m}$, $F_{\text{ext}} = 20 \text{ N}$, $\mu_k = 0.9$.

When an object is in motion on the horizontal surface, it experiences two forces.

a. External force, $F_{\text{ext}} = 20 \text{ N}$

b. Kinetic friction,

$f_k = \mu_k mg = 0.9 \times (2) \times 10 = 18 \text{ N}$.

The work done by the external force $W_{\text{ext}} = F_s = 20 \times 10 = 200 \text{ J}$

The work done by the force of kinetic friction $W_k = f_k d = (-18) \times 10 = -180 \text{ J}$

Here the negative sign implies that the force of kinetic friction is opposite to the direction of displacement.

The total work done on the object

$W_{\text{total}} = W_{\text{ext}} + W_k = 200 \text{ J} - 180 \text{ J} = 20 \text{ J}$.

Since the friction is a non-conservative force, out of 200 J given by the external force, the 180 J is lost and it

can not be recovered.

Solved Problems

Ex.3

Solved Problems

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Nonconservative Forces

- ☐ A nonconservative force is one for which work depends on the path taken.
- ☐ Friction is a good example of a nonconservative force.
- ☐ An important characteristic is that the work done by a nonconservative force adds or removes mechanical energy from a system.
- ☐ Friction, for example, creates thermal energy that dissipates, removing energy from the system.

The amount of the happy face erased depends on the path taken by the eraser between points A and B, as does the

work done against friction. Less work is done and less of the face is erased for the path in (a) than for the path in (b).

The force here is friction, and most of the work goes into thermal energy that subsequently leaves the system (the

happy face plus the eraser). The energy expended cannot be fully recovered.

3080 How Nonconservative Forces Affect Mechanical Energy

- ☐ Mechanical energy may not be conserved when nonconservative forces act.
- ☐ For example, when a car is brought to a stop by friction on level ground, it loses kinetic energy, which is dissipated as thermal energy, reducing its mechanical energy.

Comparison of the effects of conservative and nonconservative forces on the mechanical energy of a system. (a) A

system with only conservative forces. When a rock is dropped onto a spring, its mechanical energy remains constant

(neglecting air resistance) because the force in the spring is conservative. The spring can propel the rock back to its

original height, where it once again has only potential energy due to gravity. (b) A system with nonconservative forces.

When the same rock is dropped onto the ground, it is stopped by nonconservative forces that dissipate its mechanical

energy as thermal energy, sound, and surface distortion. The rock has lost mechanical energy.

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Linear Momentum and Force

Linear momentum is defined as the product of a system's mass multiplied by its velocity. In symbols, linear momentum is expressed as $p = mv$.

The SI unit for momentum is $\text{kg} \cdot \text{m/s}$

Example: (a) Calculate the momentum of a 110-kg football player running at 8.00 m/s. (b) Compare the player's

momentum with the momentum of a hardthrown 0.410-kg football that has a speed of 25.0 m/s.

Solution for (a)

To determine the momentum of the player, substitute the known values for the player's mass and speed into the

equation. $p = mv$

$$p_{\text{player}} = (110 \text{ kg})(8.00 \text{ m/s}) = 880 \text{ kg} \cdot \text{m/s}$$

Solution for (b)

To determine the momentum of the ball, substitute the known values for the ball's mass and speed into the equation.

$$p = mv$$

$$p_{\text{ball}} = (0.410 \text{ kg})(25.0 \text{ m/s}) = 10.3 \text{ kg} \cdot \text{m/s}$$

Discussion

Although the ball has greater velocity, the player has a much greater mass. Thus the momentum of the player is much

greater than the momentum of the football, as you might guess. As a result, the player's motion is only slightly affected if

he catches the ball. We shall quantify what happens in such collisions in terms of momentum in later sections.

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Momentum and Newton's Second Law

Newton stated his second law of motion in terms of momentum: The net external force equals the change in

momentum of a system divided by the time over which it changes. Using symbols, this law is

$$F_{\text{net}} = \Delta p / \Delta t ,$$

where F_{net} is the net external force, Δp is the change in momentum, and Δt is the change in time.

Linear Momentum and Force

Change in momentum Δp is given by $\Delta p = \Delta(mv)$.

If the mass of the system is constant, then $\Delta(mv) = m\Delta v$.

So that for constant mass, Newton's second law of motion becomes $F_{\text{net}} = \Delta p / \Delta t = m\Delta v / \Delta t$.

Because $\Delta v / \Delta t = a$, $\therefore F_{\text{net}} = ma$

3080 WORKED EXAMPLES

Example 1: During the in the US Open at USTA Billie Jean King National Tennis Center on September 8, 2022, Ons

Jabeur of Tunisia hit the fastest recorded serve in a premier women's match, reaching a speed of 58 m/s (209 km/h).

What is the average force exerted on the 0.057-kg tennis ball by Ons Jabeur's racquet, assuming that the ball's speed

just after impact is 58 m/s, that the initial horizontal component of the velocity before impact is negligible, and that

the ball remained in contact with the racquet for 5.0 ms (milliseconds)?

Solution

When mass is constant, the change in momentum is given by $\Delta p = m\Delta v = m(v_f - v_i)$.

To determine the change in momentum, substitute the values for the initial and final velocities into the equation,

$$\Delta p = m(v_f - v_i)$$

$$= (0.057 \text{ kg})(58 \text{ m/s} - 0 \text{ m/s})$$

$$= 3.306 \text{ kg} \cdot \text{m/s} \approx 3.3 \text{ kg} \cdot \text{m/s}$$

Now the magnitude of the net external force can be determined by using $F_{\text{net}} = \Delta p / \Delta t$:

$$F_{\text{net}} = \Delta p / \Delta t = 3.306 \text{ (kg} \cdot \text{m/s)} / 5.0 \times 10^{-3} \text{ s}$$

$$= 661 \text{ N} \approx 660 \text{ N}$$

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Conservation of Momentum

Consider what happens if the masses of two colliding objects are more similar than the masses of a football player and Earth—for example, one car bumping into another

A car of mass m_1 moving with a velocity of v_1 bumps into another car of mass m_2 and velocity v_2 that it is following. As a

result, the first car slows down to a velocity of v'_1 and the second speeds up to a velocity of v'_2 .

The momentum of each

car is changed, but the total momentum p_{tot} of the two cars is the same before and after the collision (if you assume

friction is negligible).

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Change in Momentum

$$\Delta p_1 = F_1 \Delta t,$$

$$\Delta p_2 = F_2 \Delta t$$

where F_2 is the force on car 2 due to car 1, and we assume the duration of the collision Δt is the same

for both cars.

We know from Newton's third law that $F_2 = -F_1$,

and so $\Delta p_2 = -F_1 \Delta t = -\Delta p_1$

Thus, the changes in momentum are equal and opposite, and $\Delta p_1 + \Delta p_2 = 0$.

Because the changes in momentum add to zero, the total momentum of the two-car system is constant. That is,

$p_1 + p_2 = \text{constant}$,

$p_1 + p_2 = p'_1 + p'_2$,

where p'_1 and p'_2 are the momenta of cars 1 and 2 after the collision.

In equation form, the conservation of momentum principle for an isolated system is written

$p_{\text{tot}} = \text{constant}$,

Or $p_{\text{tot}} = p'_{\text{tot}}$,

where p_{tot} is the total momentum and p'_{tot} is the total momentum some time later. An isolated system

is defined to be one for which the net external force is zero $F_{\text{net}} = 0$.

Elastic Collision

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An elastic collision is one that conserves internal kinetic energy.

CONSERVATION OF MOMENTUM

When a collision occurs in an isolated system, the total momentum of the system doesn't change with the passage of

time. Instead, it remains constant both in magnitude and in direction. The momenta of the individual objects in the

system may change, but the vector sum of all the momenta will not change. The total momentum, therefore, is said to

be conserved.

When no net external force acts on a system, the total momentum of the system remains constant in time.

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WORKED EXAMPLE

A 0.150-kg baseball, thrown with a speed of 40.0 m/s, is hit straight back at the pitcher with a speed of 50.0 m/s.

(a) What is the impulse delivered by the bat to the baseball?

(b) Find the magnitude of the average force exerted by the bat on the ball if the two are in contact for 2.00 $\times 10^{-3}$ s.

Solution (a) Find the impulse delivered to the car.

Calculate the initial and final momenta of the car:

(b) Find the average force exerted on the car.

Apply Equation 6.6, the impulse-momentum theorem:

The impulse is just the difference
between the final and initial momenta:

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CLASS WORK

1. An archer stands at rest on frictionless ice and fires a
0.500-kg arrow horizontally at 50.0 m/s.

(See Fig.6) The combined mass of the archer and bow
is 60.0 kg. With what velocity does the archer move
across the ice after firing the arrow? (Fig.6)

2. A 70.0-kg man and a 55.0-kg woman on ice skates stand
facing each other. If the woman pushes the man backwards
so that his final speed is 1.50 m/s, at what speed does she
recoil?

3. A car and a large truck traveling at the same speed collide
head-on and stick together. Which vehicle experiences the
larger change in the magnitude of its momentum? (a) the car
(b) the truck (c) the change in the magnitude of momentum is FIG. 7 (a) Before and (b) after a
perfectly
the same for both (d) impossible to determine inelastic head-on collision between two objects.

Paper 7 of 7 — PHY 101 Thermodynamics and Sound

Level 100 • First Semester • Handout

THERMODYNAMICS

>Laws of Thermodynamics:

- Zeroth Law
- First Law
- Second Law
- Third Law

>Sound

- Definition
- Classification
- Determination of speed in various media

Recommended Text Books

- Halliday D, Resnick R and Walker J (2001)
Fundamentals of Physics. John Wiley & Sons. Inc. New York.
- Young H.D., Freedman R. A (2008): University
Physics Volume 12th Edition, Sears and Zemansky's.
- Eugene Hecht (2015) College Physics, Eleventh
Edition, Schaum's Outlines

THERMODYNAMICS

- It is the study of the thermal energy or internal energy of a system and its transformation.
- The concept of heat arises from sensation of warmness or coldness which is immediately experienced on touching a body.
- This sensation does not however give a quantitative scientific measure of the state of the body with respect to heat.
- It only yields qualitative results which may vary according to external circumstances.
- Thus the central concept of thermodynamics is temperature.
- Temperature is one of the seven SI fundamental quantities and its unit is Kelvin. Other units include Celsius, Fahrenheit and Rankine.
- The general equation for converting from one scale to another is given by

$$X_o - XL \quad Y_o - YL$$

$$X_U - X_L \quad Y_U - Y_L$$

where the subscript L and U represent the lower fixed point and the upper fixed points respectively on scale of reference.

- Temperature has a distinct lower limit but apparently no upper limit.
- This limiting lower temperature is the zero of the Kelvin temperature, which is often called absolute zero.

Variation in the property of a body with change in temperature

- The properties of many bodies change as their temperature are altered. For instance water exists as solid (ice) at $T \leq 273 \text{ K}$, as liquid at $273 < T \leq 373 \text{ K}$ and as gas or vapour at $T > 373 \text{ K}$.
- That property of a body that changes linearly with temperature includes volume, length, pressure, electrical resistance and colour of filters.
- These properties that change linearly with temperatures are called thermometric fluids.
- Thus a thermoscope, that is a temperature sensing device, can be constructed using anyone of the properties listed above.
- When the instrument is calibrated then it becomes a thermometer.

Linear Variations of some basic Properties with Temperature
Property Variation with Comment
Temperature

Length $l = l_0 (1 + \alpha \Delta T)$ expansion.
 α is the linear coefficient of

$$A = A_0 (1 + \beta \Delta T)$$

β is the superficial coefficient

Area of expansion. $\beta = 2\alpha$

$$V = V_0 (1 + \gamma \Delta T)$$

γ is the cubic coefficient of

Volume expansion. $\gamma = 3\alpha$

$$\Delta P \approx$$

$$T = 273.16 \text{ K at lim}$$

$$\Delta g \text{ as } \Delta P \rightarrow 0$$

P_3 is the pressure at the triple

Pressure P_3

point of water.

α is the temperature coefficient

Electrical (

$$R = R_0 [1 + \alpha \Delta T] \text{ of resistance.}$$

Resistance

Zeroth Law of Thermodynamics

- Zeroth law of thermodynamics states that if bodies A and B are separately in thermal equilibrium with a third body C, then they are in thermal equilibrium with each other.
- This implies that when two bodies are in thermal equilibrium, their temperatures are equal.
- Thus establishing the concept of temperature as a valid concept.

Heat

- Heat is a form of energy in Transit.
- Thus when two bodies that are at different temperature come in contact, there will be a spontaneous movement of heat from the body with higher temperature to a body with lower temperature and not vice-versa except work is done on the system.

Mode of Heat Transfer

- Conduction: This is the form of heat transfer in solids. When one end of a solid is in contact with a heat source, the atoms gain thermal energy and begin to vibrate from their sites. This process causes collision with neighbouring atoms since they are held in

their lattice sites and thus heat is gradually transferred to the colder end. Materials with good thermal conductivities also have good electrical conductivities.

$X_o - X_L Y_o - Y_L$

Fourier's Law of Heat Transfer

$XU - X_L YU - YL$

Thermal Conductivities of Some Materials at Room Temperature

Material Thermal Conductivity

Wm-1K-1 Wm-1K-1

Diamond 2300 Iron 80

Silver 430 Tin 68

Copper 400 Lead 35.6

Gold 320 Glass 0.80.7

Aluminium 240 Brick

- Convection: This is the mode heat transfer in fluids i.e. liquid and gases. The layer of fluid nearest the heat source absorbed heat energy, expands and moves up to be replaced by a colder layer which in turn absorbs heat energy and the process continues until all the fluid gets heated up.

Law of conservation of Heat Energy

Newton's Law of Cooling

- The mathematical representation of the Newton's law of cooling is given by
- $T(t) = T(e) + (T(o) - T(e))e^{-kt}$
- Where $T(t)$ is the temperature of the given object at time t .
- $T(e)$ is the temperature of the environment the object is located.
- $T(o)$ is the initial temperature of the object.
- k is the decay constant
- e is the natural logarithm.
- t is the time.
- Radiation: Radiant energy is electromagnetic in

nature and does not require a medium for its transmission.

- This is the method by which Earth receives its warmth from the Sun.
- An ideal black body absorbs all radiant energy that falls on it while an ideal white body reflects all radiant energy that falls on it.
- Any object with a temperature above absolute zero radiates and absorbs radiant energy.
- When a body radiates more energy than it absorbs, then it cools down.
- When a body absorbs more energy than it radiates, then it warms up.

Stefan-Boltzmann Law of Heat

Radiation

Energy Transfer as Heat and Work

- A system can be taken from a given initial state to a given final state by an infinite number of processes.
- Heat may or may not be involved, and in general, the work W and the heat Q will have different values for different processes.
- Thus heat and work are said to path-dependent quantities.
- When a system changes from a given initial state to a given final state, both the work W and the heat Q depend on the nature of the process.

Energy Transferred as Heat and Work

Relationship between Work and Area in a P-V diagram

- The work done by a fluid in an expansion is equal to the area beneath the expansion curve on a P-V diagram.
- In a cyclic process, the work output per cycle done by a fluid is equal to the area enclosed by the P-V diagram representing the cycle.

the P-V diagram.

P P P

V V V

Fig. 2a Fig. 2b Fig. 2c

P P P

V V V

Fig. 2d Fig. 2e Fig. 2f

- Experimentally, however, the quantity $Q-W$ is the same for all processes.
- It depends only on the initial and final states and does not depend at all on how the system gets from one to the other.
- All other combinations of Q and W , including Q alone, W alone, $Q+W$ and $Q-2W$ are path dependent.
- The quantity $Q-W$ represents a change in some intrinsic property of the system. This property is called the internal energy, U .
- Thus $\Delta U = U_f - U_i = Q - W$

First Law of Thermodynamics

- Hence the first law of thermodynamics states that if an amount of heat flows into a system, then this energy must appear as increased internal energy for the system and / or work done by the system on its surroundings.

- This can be written as

$$\Delta Q = \Delta U + \Delta W$$

- The law is a statement of the law of conservation of energy.

Thermodynamic Processes

- **Adiabatic** : This is one that occurs so rapidly or occurs in a system that is so well insulated that no transfer of energy as heat occurs between the system and its surrounding.
- For this system $\Delta Q = 0$
- In adiabatic process, the first law becomes $0 = \Delta U + \Delta W$
- Any work done by the system is done at the expense of the internal energy.
- Any work done on the system serves to increase the internal energy.
- For an ideal gas changing from conditions $(P_1V_1T_1)$ to $(P_2V_2T_2)$ in an adiabatic process

then

$$\gamma = \frac{C_p}{C_v} = \frac{f+2}{f}$$

$$P_1 V_1^\gamma = P_2 V_2^\gamma \text{ and } T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\gamma = \frac{C_p}{C_v}$$

where $\gamma = \frac{C_p}{C_v}$

C_p

C_v

is the ratio of specific heat at constant pressure to that at constant volume.

- When a gas is heated at constant volume, the heat supplied goes to increase the internal energy of the gas molecules.
- However, when a gas is heated at constant pressure, the heat supplied not only increases the internal energy of the molecules but also does mechanical work in expanding the gas against the opposing constant pressure.
- Hence the specific heat of a gas at constant pressure C_p is greater than its specific heat at constant volume, C_v . For an ideal gas of molecular mass M ,

Where $R = 8314 \text{ J/Kmol.K}$ is the universal gas constant thus C_p is in J/kg.K or $\text{J/kg}^\circ\text{C}$.

The specific heat ratio is greater than unity for a gas. The kinetic theory of gases indicates that for monatomic gases (such as He, Ne, Ar), $\gamma = 1.67$. For diatomic gases (such as O_2 , H_2), $\gamma = 1.40$ at ordinary temperatures.

$$\Delta Q = \Delta U$$

Isobaric: This is a process that is carried out at a constant pressure.

Isovolumic: This is a process carried out at a constant volume.

When a gas undergoes such a process $\Delta W = P\Delta V = 0$, hence the first law of thermodynamics becomes,

Isothermal Process: This is one carried out under constant temperature.

In the case of an ideal gas, where constituent atoms or molecules do not interact when separated, $\Delta U = 0$ in an

isothermal process. However, this is not true for many other systems. For example $\Delta U \neq 0$ as ice melts to water at 0°C even though the process is isothermal.

For an ideal gas, $\Delta U = 0$ in an isothermal change and so the first law becomes $\Delta Q = \Delta W$

For an ideal gas changing isothermally from (P_1V_1) to (P_2V_2) , then

$$P_1 V_1 = P_2 V_2$$

$$\Delta Q = \Delta W = P_1 V_1 \ln \frac{P_2}{P_1} = 2.30 P_1 V_1 \log \frac{P_2}{P_1}$$

$$P_1 V_1 = P_2 V_2$$

where $\ln$ and $\log$ are logarithms to base e and base 10 respectively.

The Efficiency of a Heat Engine

This is defined as:

work output

eff =

heat input

The Carnot cycle depicts the most efficient cycle possible for a heat engine. An engine that operates in accordance to this cycle between a hot reservoir and a cold reservoir, T_h and T_c respectively has efficiency

T_c

$$\text{eff max} = 1 - \frac{T_c}{T_h}$$

T_h

6.11

where the temperature is to be quoted in Kelvins.

Second Law of Thermodynamics

The second law of thermodynamics is defined in three equivalent ways:

- When two bodies at different temperatures are in contact, heat flows spontaneously from a hotter to a colder object, but not vice versa, except work is done.
- There is no heat engine that cycle continuously that can change all its heat-input to useful work-output.
- If a system undergoes spontaneous change, it will change in such a way that its entropy will increase or, at best remains constant.

Entropy (S)

- This is a state variable for a system in equilibrium.

For instance S is always the same for the system when it is in a given equilibrium state.

- Like P, V and U, the entropy is a characteristic of the system at equilibrium.
- When heat ΔQ enters a system at an absolute temperature T, the resulting change in entropy of the system is

$$\Delta Q$$

$$\Delta S = \frac{\Delta Q}{T}$$

Provided the system changes
in a reversible way

Alternative Definition of Entropy

- Another fully equivalent definition of entropy can be given from a detailed molecular analysis of the system.
- It states that if a system can achieve a particular state (for instance particular values of P,V,T and U) in Ω different ways (i.e different arrangements of the molecules) then the entropy of the state is

$$S = K_B \ln \Omega$$

Where K_B is the Boltzmann's
constant = $1.38 \times 10^{-23} \text{ JK}^{-1}$

Third Law

- The third law of thermodynamics states that absolute zero cannot be attained by any procedure in a number of steps.
- Absolute zero can be approached closely but it can never be reached.

Sound

- It is a longitudinal compression wave which requires a material medium for its transmission.
- There are three classes of sounds.

Classification of Sound

Class Frequency Range Application

Infrasonic $< 20 \text{ Hz}$ Appears as a vibration which is detected by some animals before the occurrence of a land tremor or volcanic eruption.

Audible 20 Hz to 20 KHz Communication and entertainment

Ultrasonic > 20 KHz Medical diagnostic imaging
Ultrasonic machining using high frequency low amplitude vibrations to create holes and other cavities.

Ocean depth sounding

Determination of the Speeds of Sound in Various Media

- i In an Ideal Gas
- In an ideal gas of molecular mass M and absolute temperature T , sound has a speed given by
- Where R is the gas constant
- li. In Solids
- The speed, s of sound in solids is given by:

For a solid bar, the modulus is the Young modulus while for a liquid it is the bulk modulus.

Thus for a solid bar:

- For a liquids, where B is the bulk modulus, the speed is given by:
- The minus sign cancels the negative numerical value Δv

Speeds of Sound at Different Temperature

- The speed of sound in air at 0°C is 331 m/s.
- The speed of sound is independent pressure, frequency and wavelength.
- Two speeds of sounds v_1 and v_2 at absolute temperatures T_1 and T_2 respectively are related by:
- Thanks for Your Rapt Attention